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Comparative analysis of interfacial characteristics and jetting phenomena in explosive, vaporizing foil actuator, and laser impact welding

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Abstract

This study compares three high-velocity impact welding techniques (explosive welding (EXW), vaporizing foil actuator welding (VFAW), and laser impact welding (LIW)), using AA1100-O flyers and AISI 1018 targets over flyer thicknesses of 0.07 mm-57.15 mm and impact velocities of 510 m/s-750 m/s. Photon Doppler velocimetry (PDV) quantified flyer kinematics, scanning electron microscopy (SEM) resolved interfacial morphology, and smoothed particle hydrodynamics (SPH) captured pressure wave propagation, collision-induced heating and jetting. Interfacial wavelengths spanned ~6 µm (LIW), ~37 µm (VFAW), to ~2.9 mm (EXW). In EXW, wave size increased with flyer thickness (6.35→22.86 mm) despite decreasing velocity (750→400 m/s), indicating that flyer mass/kinetic energy can dominate over velocity alone. Melting layer width in EXW rose from ~1.3 mm to ~3.1 mm and correlated with thicker IMCs (~3–5 μm → ~20–30 μm) localized at wave crests, while troughs remained largely IMC-free. Across LIW→VFAW, peak interface pressures decreased (e.g., LIW ~5.2×10^7→2.64×10^7 kPa, VFAW ~1.23×10^7→3.77×10^6 kPa with increasing flyer thickness), producing longer wavelengths and broader but cooler thermal zones. Jetting exhibited a decoupling, quantity increased with flyer thickness/spot footprint (EXW>VFAW>LIW), whereas velocity tracked peak pressure (LIW>VFAW>EXW). Collectively, the results show that pressure-pulse duration and spatial distribution, not velocity alone govern interfacial waviness, thermal gradients, and jetting. The unified pressure-wave framework provides process design guidance, EXW for large, robust clads, VFAW for medium-thickness lightweight joints and LIW for fine-scale joining.

Keywords

Explosive welding (EXW); Vaporizing foil actuator welding (VFAW); Laser impact welding (LIW); Smoothed particle hydrodynamics (SPH); Interface prediction; Jetting in impact welding

1. Introduction

Impact welding is a solid-state joining process used for various material combinations that utilizes a high-speed collision of metals to remove surface oxides and directly bond the bare metals via pressure-driven metallic contact [1], [2], [3]. Its minimal thermal input enables joining a wide range of similar and dissimilar materials with minimal microstructural modification [4], [5], [6]. Reported combinations include ferrous alloys, aluminum, titanium, copper, magnesium, tantalum, niobium, bulk metallic glass, and high-entropy alloys [1], [2], [3], [4], [5], [6], [7], [8], [9], [10].

Several impact welding techniques have been developed to cover different thickness regimes. Explosive welding (EXW) is widely used for thick-section applications (>5 mm) such as pressure vessels and cladded structural materials [11]. Its interfacial waves are macroscopic and influenced by the collision point velocity V_c, which depends on detonation speed and standoff distance [12],

[13]. Vaporizing foil actuator welding (VFAW), targets intermediate thicknesses (0.2–5 mm) [14] and has been used for automotive dissimilar joining [15], [16], [17] since its introduction in 2013 by Vivek et al. [15]. Its waveforms are similar to EXW but smaller in amplitude (tens to hundreds of microns) [18]. Laser impact welding (LIW) is used for ultra-thin foils (20–250 µm) [19] and generates pressure via pulsed lasers of nanosecond duration [20], [21].

The formation mechanisms of impact-welded interfaces have been extensively studied, primarily through hydrodynamic instability theory [22] and plastic flow models [23]. Heat generation from plastic deformation has been identified as a key driver for intermetallic compound (IMC) formation in dissimilar systems [24]. It is strongly influenced by parameters such as impact velocity and collision angle [11], [12], as well as flyer and target thickness [25], [26] and material combination [20]. While each process differs in pressure generation, chemical detonation in EXW [7], [11], vaporized foil-induced plasma in VFAW, and laser-induced plasma in LIW [19], [21], the underlying dynamics of oblique collision, pressure propagation, and jetting remain common to all. Across these processes, the pressure-pulse waveform, characterized by peak magnitude, duration, and spatial spread, governs interface morphology more fundamentally than nominal impact velocity as priorly studied [27], [28], [29] link these pulse parameters to wave growth, IMC formation, and jetting behavior across high-velocity welding methods.

This study systematically compares the interfacial morphology and jetting behavior of EXW, VFAW, and LIW through experiments and smoothed particle hydrodynamics (SPH) simulations. Rather than evaluating performance, the focus is on understanding how pressure pulse waveform characteristics, namely magnitude, duration, and spatial spread, dictate interfacial and jetting phenomena. While previous studies have examined pulse duration and peak pressure effects in isolation for EXW [30], VFAW [31], and LIW [32], a comprehensive comparison across all three processes is lacking. This work develops a unified pressure-driven collision mechanics framework to offer predictive insight into impact welding mechanisms across scales.

2. Experimental methods

Experiments were conducted for all three welding processes (EXW, VFAW and LIW) using, AA1100-O as the flyer material and steel 1018 (mild steel) as the target. The materials were chosen because aluminum alloys and low carbon steel are both widely available structural materials, and the joining of the two is of great interest to many industries [11], [12]. The impact velocity of each process was monitored and controlled by photon Doppler velocimetry (PDV), which was made available for characterization of high-strain rate processing of metals [33], [34]. The impact velocity measurement and weld production were carried out independently.

EXW experiments were carried out in a controlled environment at NobelClad™, with a 9.32 mm standoff tubing maintaining the gap between the AA1100 flyer and 1018 steel target. The flyer plate, 622.3 mm long, 381 mm wide, and with thicknesses of 6.35 mm, 9.53 mm, 25.40 mm, and 57.15 mm, was launched by explosive detonation. The velocity profiles of each thickness were measured post-welding using PDV as, after welding, the plates were re-launched by explosive detonation, and the velocity profile of each thickness was obtained by PDV. The impact velocity was compared with the weld that was produced with the same input process parameters. The experimental setup used for velocity measurement of explosive welding is shown in Fig. 1(a).

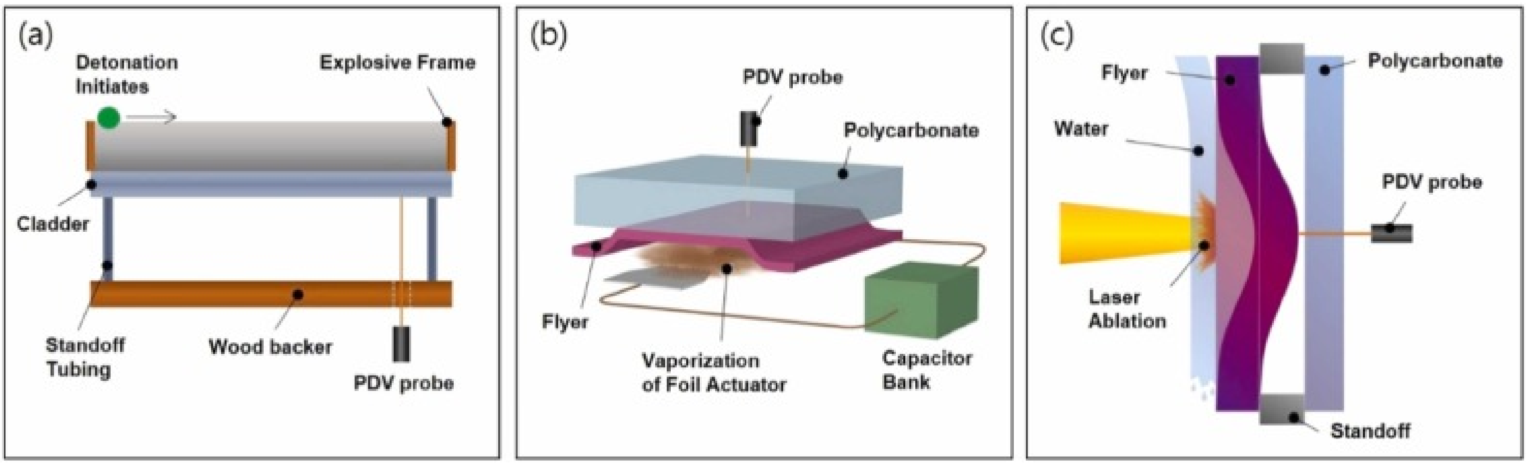

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Fig. 1. Schematics showing impact velocity measurement setup for (a) EXW, (b) VFAW, and (c) LIW.

VFAW experiments as well were performed for 6.35 mm of AA1100 flyer for comparison with EXW. As Lee et al. had discussed before about the applicability of VFAW for thick (over 6 mm) flyer plates [18]. For velocity measurement, the VFAW experiment was conducted with a transparent polycarbonate target to allow the PDV probe to measure the velocity without getting damaged as shown in Fig. 1(b). A standoff distance of 1.6 mm or 2.4 mm was utilized, depending on whether the flyer thickness is less or greater than 1 mm. The selection of the standoff distance was based on trials in achieving successful welds. Furthermore, two sets of impact velocity measurements were performed; first, the input energy was adjusted to match a single impact velocity (670 m/s in this case), while the flyer thickness was varied between 0.127 and 1.016 mm for comparison with the LIW. For the second set of experiments, the input energy was kept constant at 5 kJ while the flyer thickness was varied between 0.5 and 1 mm. The velocity profile of each flyer thickness was recorded.

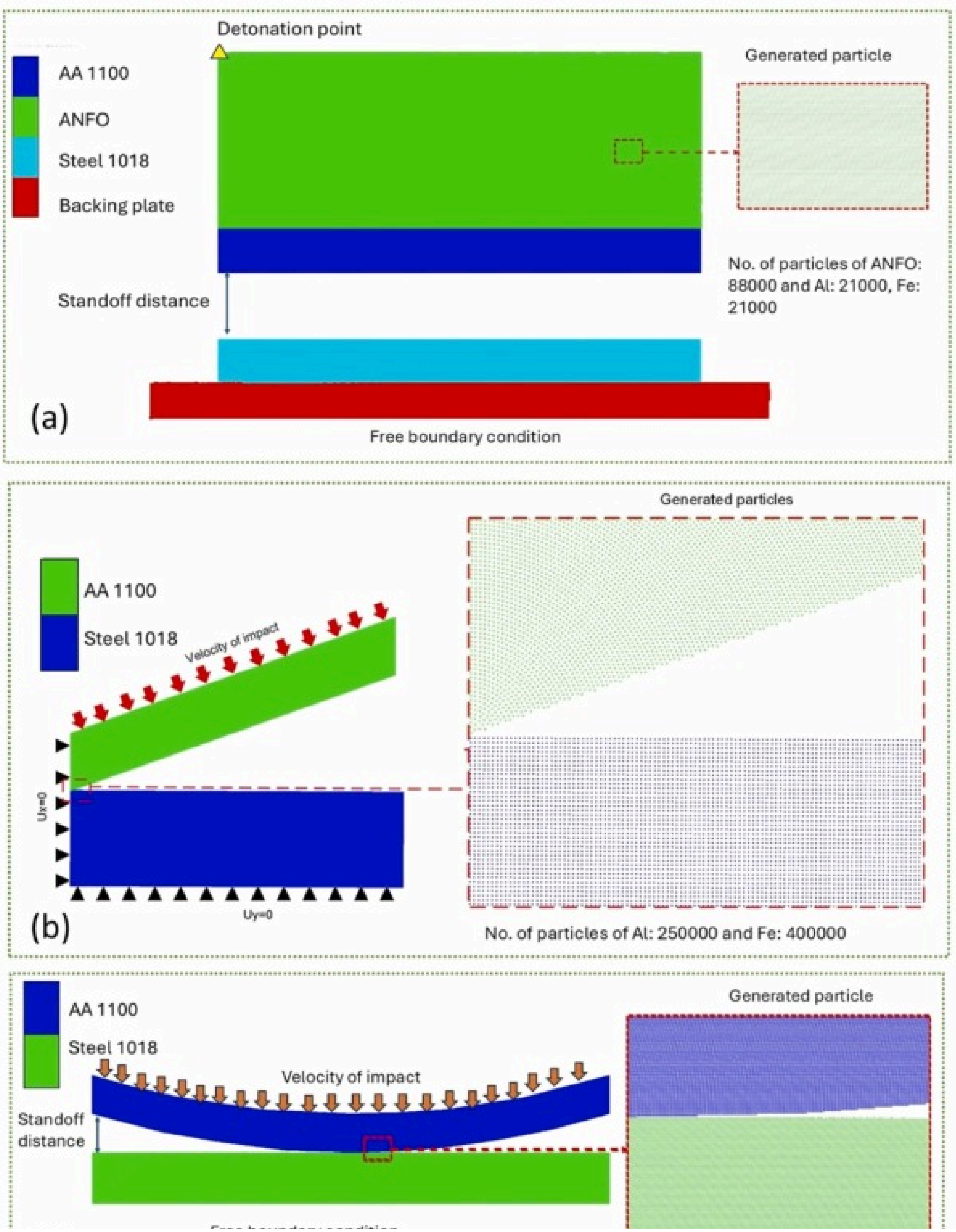
For LIW experiments, a thin AA1100 flyer of thicknesses between 0.076 and 0.152 mm was used, and samples were welded using a 3 J Continuum laser with an 8 ns pulse width at a rate of 10 Hz, producing a power density of 7.6 $\mathrm{GW} / \mathrm{cm}^2$ for 2 mm spot size. During velocity measurement, a polycarbonate sheet was placed as a target, similar to during VFAW, so that the velocity can be measured on the other side of the laser ablation, as shown in Fig. 1(c). The standoff distance was the same as each flyer thickness, and flowing water was used to confine the plasma generated from the surface ablation. After welding, cross-sectional samples were prepared and polished for metallography. The microscopic images were then compared with the velocity profiles of each impact welding process and various flyer thicknesses. The Thermo Scientific Apreo scanning electron microscope was used for the microscopic analysis.

3. Finite element modelling

Smoothed particle hydrodynamics (SPH), a mesh-free computational technique, was selected due to its robust capabilities in handling the large deformations, high strain rates, and complex material interactions characteristic of these processes [2], [35]. SPH models discretize the domain into particles, each carrying physical properties such as mass, momentum, and energy. These particles interact with their neighbors through smoothing functions, enabling the accurate simulation of highly dynamic processes [2]. This approach eliminates mesh distortion issues associated with conventional FEM, making it particularly suitable for simulating extreme conditions in impact welding [36]. Each method utilizes a two-dimensional (2D) SPH approach to capture the complex interactions and dynamics at the interfaces during the welding process.

In all SPH models, the flyer and target plates were assumed to have perfectly clean, oxide-free, and smooth surfaces, with no initial gaps at the interface. Free boundary conditions were applied at all outer edges to avoid artificial stress wave reflections, while the symmetry of the setup was preserved in the 2D domain [37]. For contact interaction, the SPH formulation automatically handles separation and re-contact without artificial constraints, and no friction was included to isolate the role of impact dynamics. Heat conduction was neglected because the simulated timescales (tens of nanoseconds to microseconds) are too short for significant diffusion. These assumptions ensure that the simulations isolate the influence of process-specific pressure pulse waveform characteristics on interface morphology and jetting behavior.

EXW model utilizes a reduced-scale modelling approach to significantly decrease computation time without affecting the analysis, as shown in Fig. 2(a). The detonation ratio was adjusted to match the experimental PDV results, and the detonator was installed on the surface edge of the explosive layer. VFAW model also utilizes a small portion of the mechanical system (Fig. 1(b)) to develop the 2D model, as shown in Fig. 2(b). During LIW, the pressure from the expanding plasma shock wave exhibits a Gaussian distribution on the flyer plate and develops a downward bulge. Therefore, to model this, the flyer plate is designed as an ideal arc, and a 2D plane model is created, as shown in Fig. 2(c), as also followed in many previous works [36], [38]. The flyer movement is governed by assigning an initial velocity with uniform distribution to the flyer material at the point of impact during both VFAW and LIW models, with the values matching the experimental PDV measurement.



Detonation point
AA 1100
ANFO
Steel 1018
Backing plate
Generated particle
Standoff distance
No. of particles of ANFO: 88000 and Al: 21000, Fe: 21000
(a)
Free boundary condition
Generated particles
AA 1100
Steel 1018
Velocity of impact
Ux=0
Uy=0
(b)
No. of particles of Al: 250000 and Fe: 400000
AA 1100
Steel 1018
Generated particle
Velocity of impact
Standoff distance

Fig. 2. 2D finite element modeling arrangement for (a) EXW, (b) VFAW and (c) LIW process.

After defining each process geometry and loading, particle spacing was selected via convergence studies to resolve jetting and interface waveforms while balancing computational cost. Constant individual particle sizes of 20 µm, 2 µm, and 1 µm were used for EXW, VFAW, and LIW models, respectively [39]. For all three processes, consistent material definitions for flyer and target plates (AA1100 aluminum and 1018 steel) were used to ensure comparability. Process parameters were based on experimental measurements The Mie–Grüneisen equation of state (EOS), which is based on the Shock Hugoniot relationship, widely used to describe the thermodynamic behavior of materials subjected to shock waves, was utilized for all the models as follows,

$$p = p_h + \Gamma \rho_0 (e - e_h),$$

and the Hugoniot pressure relationship is expressed as,

$$p_h = \rho_0 c_0^2 \frac{(1-\eta)}{(1-S\eta)^2}.$$

Here, p is pressure in the shocked material, p_h is pressure along the Hugoniot curve, Γ is Grüneisen parameter, ρ_0 is initial density of the material, e is the specific internal energy of the material, e_h is specific internal energy along the Hugoniot curve, c_0 is the Sound speed in the material at zero pressure, η is compression defined as $\eta = 1 - \frac{\rho_0}{\rho}$, where ρ is the shocked density. Furthermore, S is material constant in the linear relationship of $u_s = c_0 + Su_p$. Here, u_s and u_p are shock velocity and particle velocity [13]. EOS constants are shown in Table 1. The material behavior in all three processes was modelled using a Johnson-Cook constitutive relation as shown,

$$\sigma_{eq} = \quad (A + B\varepsilon^n)\left(1 + C\ln\left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}_o}\right)\right)\left(1 - \left(\frac{T-T_r}{T_m-T_r}\right)^m\right).$$

Table 1. Mie–Grüneisen EOS parameters for AA1100 and 1018 steel [40], [41].

Material	Γ	c_0 **(m/s)**	S_1	S_2
AA1100	1.97	5240	1.4	0.0
1018 steel	2.17	4596	1.49	0.0

Here, of a material as a function of plastic strain, strain rate, and temperature. *A* is the yield strength at reference conditions (room temperature and quasi-static strain rate), and *B* and *n* define the strain hardening behaviour, where *B* is the hardening modulus and *n* is the strain-hardening exponent. The constant *C* represents strain rate sensitivity, capturing how stress changes with increasing strain rate relative to a reference strain rate $\dot{\varepsilon}_0$, typically 1 s^{-1}. The thermal softening exponent *m* governs the reduction in stress as temperature increases from the reference temperature T_r to the melting temperature of the material T_m. ε is the equivalent plastic strain, and $\dot{\varepsilon}$ is the plastic strain rate. Together, these parameters allow the J-C model to predict material behaviour under large strains, high strain rates, and elevated temperatures. J-C constants are shown in Table 2. The Jones-Wilkins-Lee (JWL) EOS was also used to model ammonium nitrate fuel oil (ANFO) to calculate the expansion of detonation reaction products from the chemical equilibrium Chapman–Jouguet (C-J) state to large volumes as shown,

$$P = A_1\left(1 - \frac{\omega}{R_1 V}\right)e^{-R_1 V} + B_1\left(1 - \frac{\omega}{R_2 V}\right)e^{-R_2 V} + \frac{\omega E_0}{V}.$$

Where *P* is the pressure, *V* is the relative volume, E_O is the initial specific internal energy and A_1, *B*, R_2, R_2, and ω are constants of the explosive. C-J equation coefficient values are shown in Table 3. The model directly incorporated thermal softening via the J-C constitutive formulation, in which the flow stress decreases with increasing temperature. Adiabatic heating was implicitly included, as plastic and compressive work increased the E_0 in the shock EOS, which in turn elevated the local temperature and affected the J-C softening term. This approach ensured a consistent thermo-mechanical coupling without requiring an explicit thermal diffusion analysis. Furthermore, the solver applied wrap-up criteria, including a cycle limit of 10,000,000 cycles, end time, and an energy fraction threshold of 0.1, to automatically adjust the timestep size [2].

Table 2. J-C constitutive model constants for AA1100 and 1018 steel [31], [32].

Material	***A*** **(MPa)**	***B*** **(MPa)**	n	c	m	T_m **(K)**	T_r **(K)**
AA1100	148	361	0.183	0.001	0.859	933	298
1018 steel	532	229	0.302	0.027	1	1473	298

Table 3. C-J equation coefficients for ANFO [13].

V	ρ_0	E_0 **(J/Kg)**	A_1 **(GPa)**	B_1 **(GPa)**	R_1	R_2	ω
4160	0.933	2.48	49.46	1.89	3.91	1.12	0.33

4. Results and discussion

4.1. Flyer velocity evolution and pressure generation across EXW, VFAW, and LIW

While the high-speed collision of metals is necessary for any impact welding, each impact welding process uses a unique way of pressure generation. The difference in duration and type of pressure can change the behavior of the launched flyer. The first role of the generated pressure is to form the flyer material at a high strain rate so that the flyer and target materials can form an appropriate collision angle during the impact [13]. Considering the heterogeneous structure of the impact weld interface, the velocity of the flyer at the time of impact may differ depending on the location of the specific observation point within the weld zone [42]. For EXW, a location about 400 mm away from the detonator was selected to observe the microstructure and flyer velocity, whereas the velocity of VFAW and LIW was measured from the center of the flyer. Fig. 3 shows the velocity profiles of the AA1100 flyer launched by explosive detonation, vaporizing foil actuator and a laser pulse during free-flight without a target. The distance corresponds to the actual separation traversed by the flyer before collision, enabling direct correlation of acceleration behavior with standoff distance. In EXW, the 6.35 mm thick flyer exhibits a gradual acceleration and smooth velocity profile, reflecting the sustained, long-duration pressure pulse generated by explosive detonation as also elaborated by Sun et al. [13]. In contrast, the VFAW flyer of the same thickness displays noticeable velocity fluctuations attributed to internal stress waves reflecting within the thicker flyer material also mentioned by Nassiri et al. [36] As flyer thickness decreases (e.g., 0.25 mm in VFAW), the influence of stress waves diminishes, resulting in more linear acceleration driven by electromagnetic interactions. LIW, employing a 0.15 mm foil flyer, shows a much sharper acceleration over a very short travel distance (~0.15 mm), indicative of a rapid, short-duration pressure impulse. These differences in acceleration behavior across processes directly reflect variations in pressure pulse duration and spatial distribution, fundamentally influencing collision dynamics and interfacial phenomena. This linkage between waveform characteristics and interface evolution is consistent with earlier work in explosive, electromagnetic, and laser-driven impact welding [13], [22], [28], [29], [36], where both experimental and modeling approaches (including plasma pressure models) highlight the role of pulse magnitude, rise time, and decay in shaping weld interfaces.

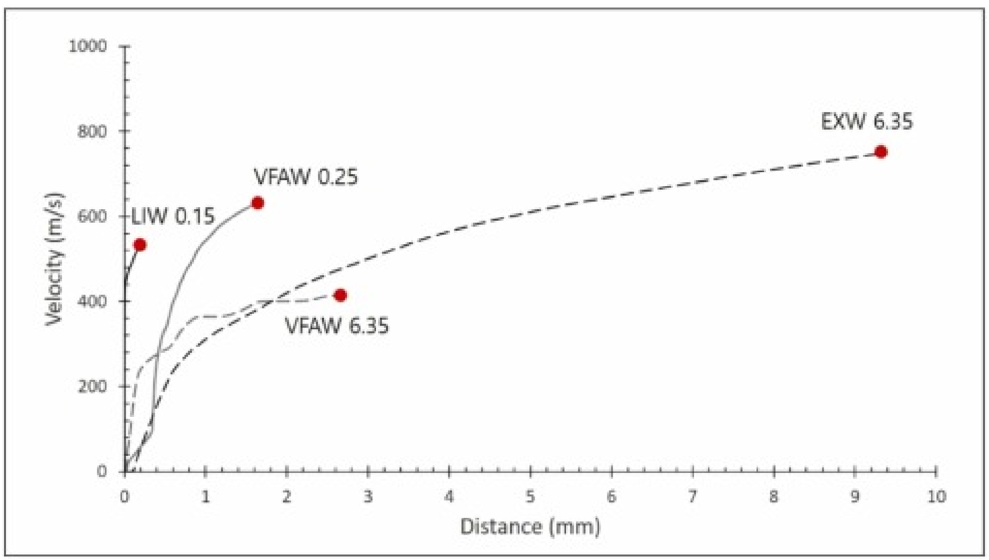


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Fig. 3. Flyer velocity profiles of EXW, VFAW and LIW processes as obtained by the PDV probe for flyer plates of different thickness during free-flight without a target in the standoff direction.

4.2. Interface morphology and waviness formation in EXW

In the EXW setup, flyer thicknesses of 6.35, 9.53, 25.40, and 57.15 mm produced impact velocities of ~750, 670, 400, and 180 m/s, respectively (Fig. 4). Welding was achieved across all conditions, but interface morphology varied significantly (Fig. 5). Wave amplitude and wavelength increased with flyer thickness despite the decrease in impact velocity, contrasting previous studies that reported a positive correlation between velocity and wave size [25]. This indicates that flyer mass and hence total kinetic energy can override velocity effects alone in controlling wave formation, and both variables must be considered when predicting interface morphology.

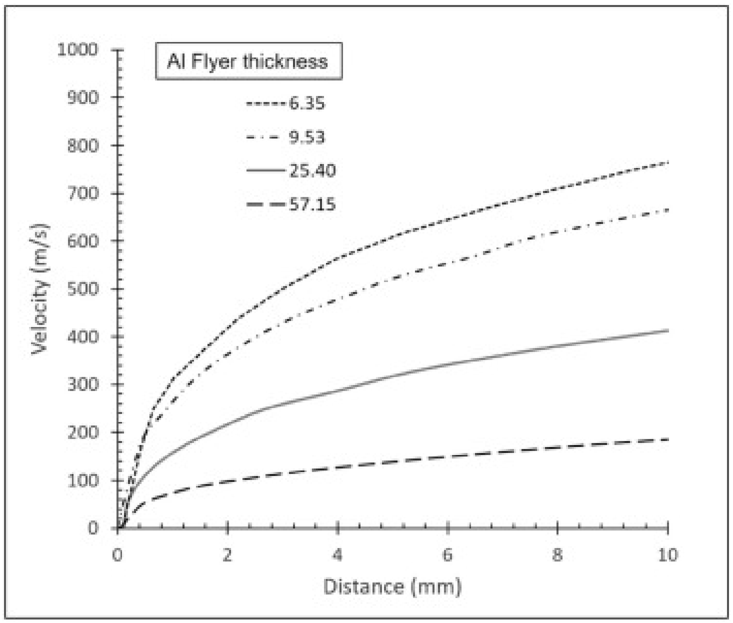

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Fig. 4. Velocity profiles with distance in the direction of standoff during free flight without target for various AA1100 flyer thicknesses during EXW.

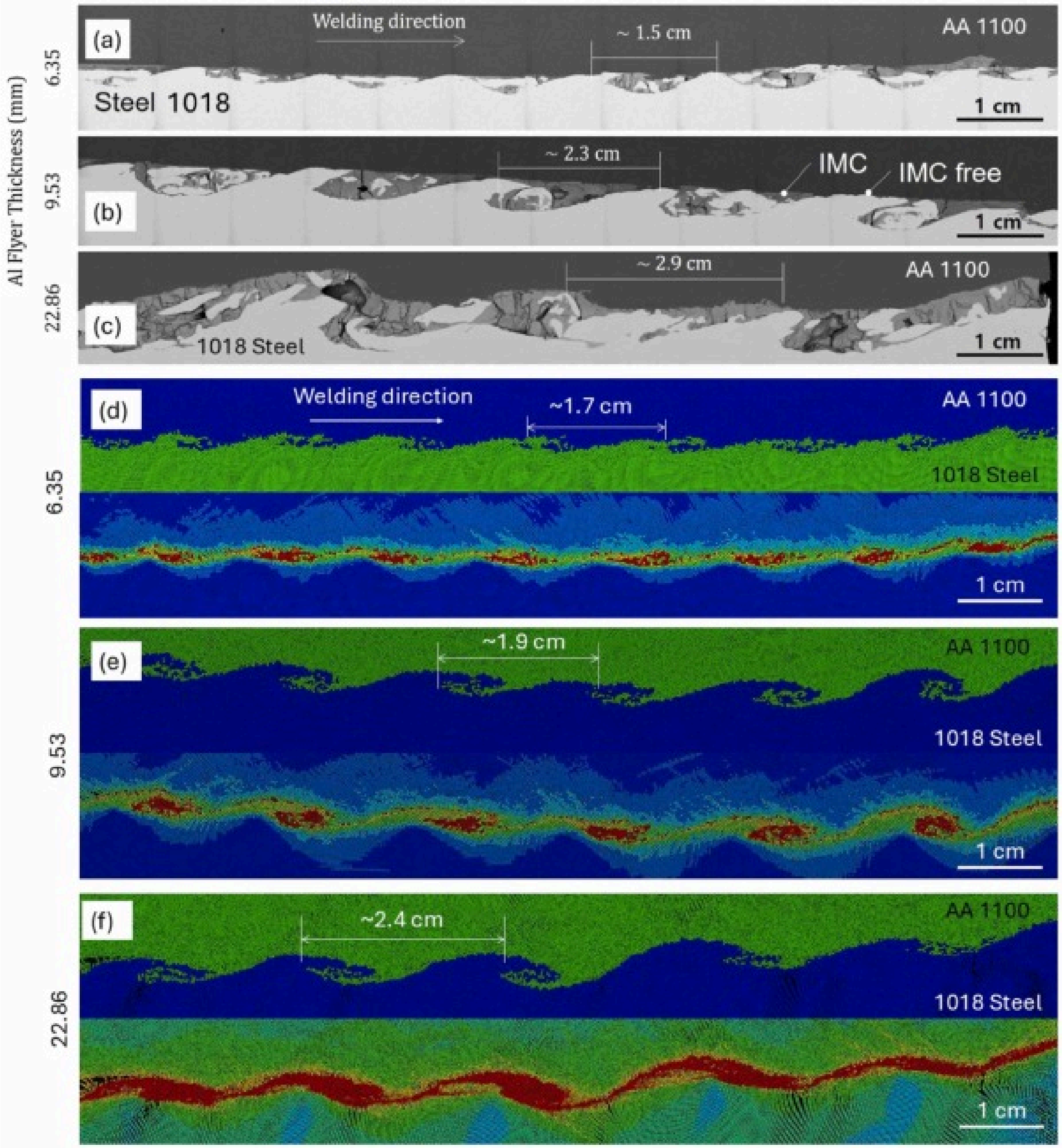

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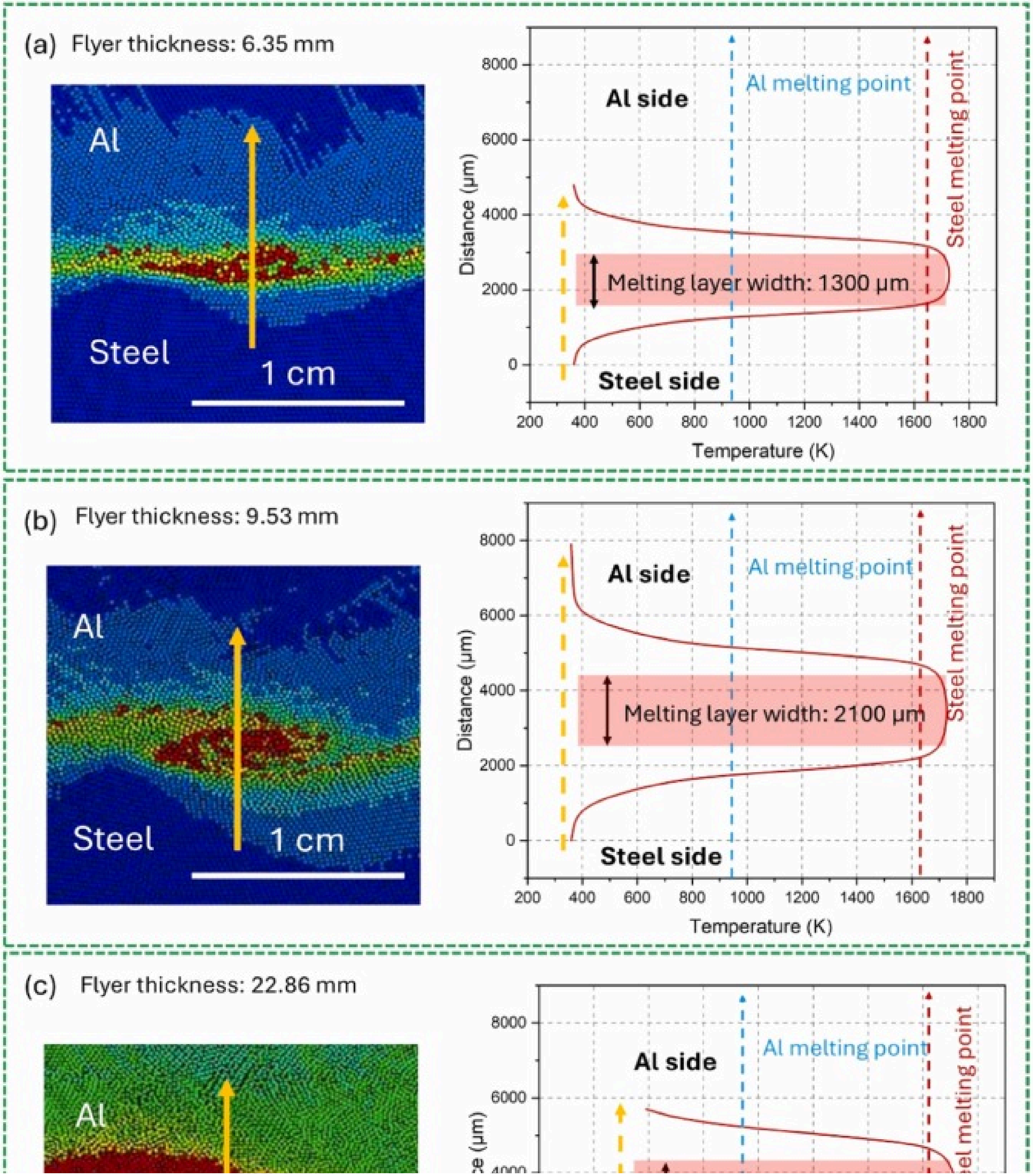
Fig. 5. Interface morphology for EXW AA1100–1018 steel produced by SEM (a) 6.35 mm flyer at 750 m/s, (b) 9.53 mm at 670 m/s, (c) 22.86 mm at 400 m/s and respectively produced by SPH (d) 6.35 mm at 750 m/s, (e) 9.53 mm at 670 m/s, (f) 22.86 mm at 400 m/s.

The average interfacial wavelengths for 6.35 mm, 9.53 mm, and 22.86 mm thick flyer plates were observed to be 1.5 mm, 2.3 mm, and 2.9 mm in the SEM images and 1.7 mm, 1.9 mm and 2.4 mm during SPH modeling, respectively. While the SPH simulations qualitatively capture major trends in interfacial morphology, the absolute quantitative agreement with experimental measurements is limited. This discrepancy stems primarily from two sources, (i) sensitivity to particle resolution and smoothing length, which influences the sharpness and fidelity of captured interface features, and (ii) reliance on simplified material models (J-C and Mie–Grüneisen EOS), which may not fully account for dynamic softening, microstructural gradients, or phase transformations at extreme strain rates as also previously stated [13], [36], [41]. Consequently, SPH results can be interpreted as semi-quantitative approximations rather than precise predictors. The wavelengths were observed to be in a nonlinear relationship with the flyer thickness and impact velocity, but it seemed to be affected more by the flyer thickness, as previously confirmed [25].

Intermetallic compound (IMC) formation was observed to be intensified with flyer thickness. At 6.35 mm, IMCs were thin (~3–5 μm) and confined to isolated wave crests, while troughs remained IMC-free, indicating predominantly solid-state (melting-free) bonding. At 9.53 mm, IMCs thickened to ~10–15 μm and became semi-continuous. At 22.86 mm, IMCs reached ~20–30 μm and were continuous across crests, indicating localized melting [11]. This spatial heterogeneity, continuous IMCs at crests versus solid-state bonding at troughs, implies that localized thermal gradients likely govern IMC continuity, rather than wave geometry alone [11], [41]. Similar crest–trough thermal asymmetry and its control over IMC formation have been reported in prior impact welding studies [25], [33], [40]. The observed microstructural variations can be further clarified by quantifying melting layer widths at the interface.

4.3. Temperature distribution and melting behavior in EXW

Fig. 6 presents temperature distributions across the EXW interface in melting-dominated regions, where localized plastic deformation and heat generation are highest. As flyer thickness increased from 6.35 mm to 22.86 mm, peak temperatures rose from slightly above the melting point of aluminum (~660°C) to > 800°C. The melting layer width increased accordingly from ~1300 μm to ~3100 μm, closely matching the regions where continuous IMC layers (20–30 μm) were observed in Fig. 5(c).


(a) Flyer thickness: 6.35 mm
Al
Steel
1 cm
Al side
Al melting point
Steel melting point
Melting layer width: 1300 μm
Steel side
Distance (μm)
Temperature (K)
(b) Flyer thickness: 9.53 mm
Al
Steel
1 cm
Al side
Al melting point
Steel melting point
Melting layer width: 2100 μm
Steel side
Distance (μm)
Temperature (K)
(c) Flyer thickness: 22.86 mm
Al
Al side
Al melting point

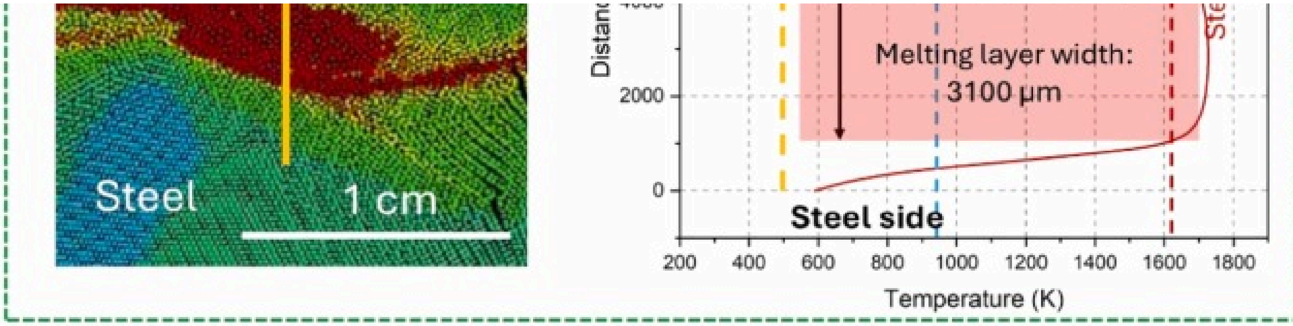


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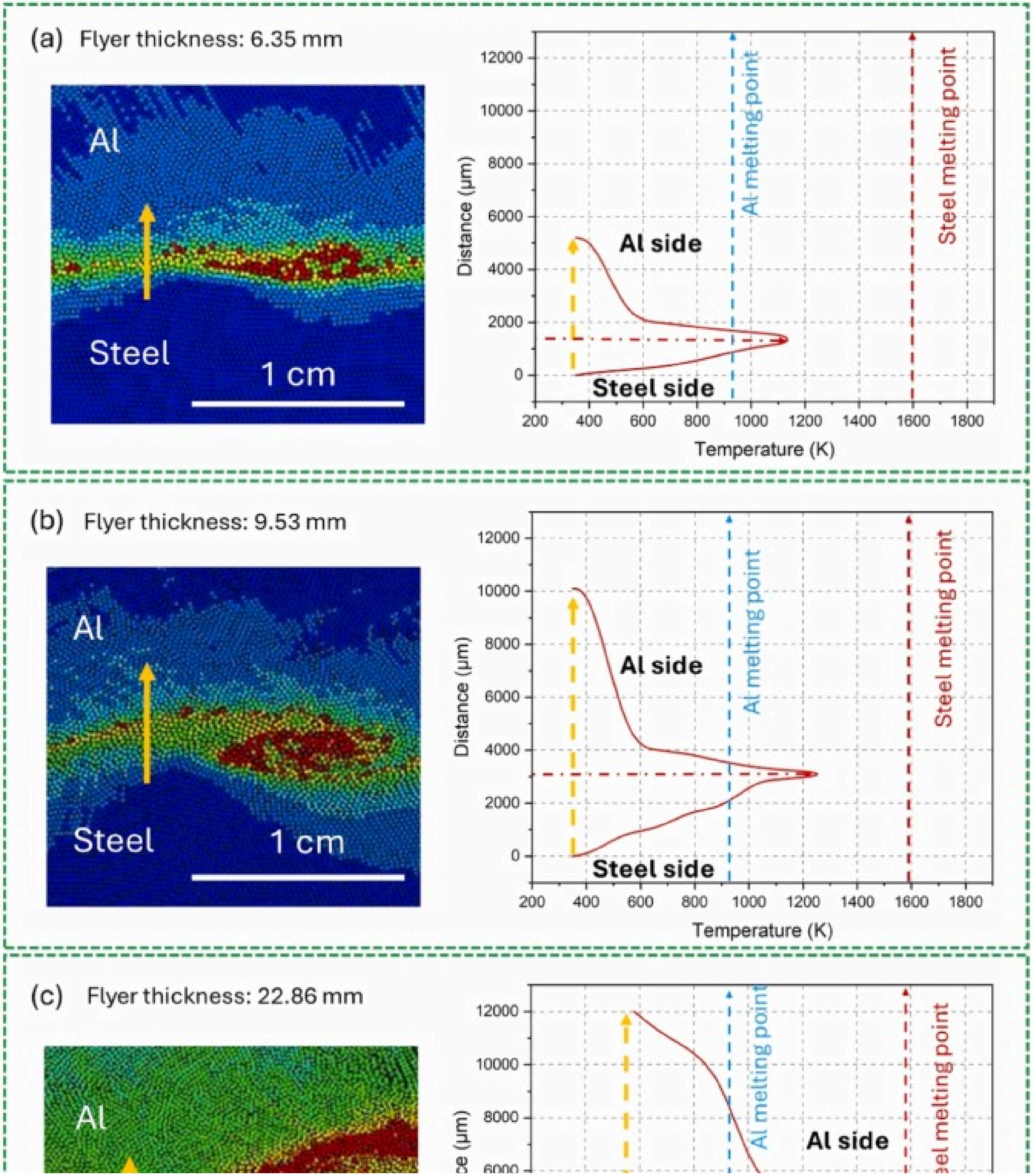
Fig. 6. EXW AA1100–1018 steel temperature distribution in melting-dominated region for (a) 6.35 mm flyer at 750 m/s, (b) 9.53 mm at 670 m/s, (c) 22.86 mm at 400 m/s.

To fully understand the spatial heterogeneity in IMC formation, temperature profiles were also evaluated in least-melting regions, shown in Fig. 6. Prior studies have reported crest–trough thermal asymmetry as a key factor influencing interfacial compound formation [24], [42]. By examining both the high-temperature crests and the minimally affected troughs, the analysis clarifies how localized thermal gradients control IMC continuity. This dual-region comparison ensures predictive accuracy and spatial resolution in interpreting bonding mechanisms.

In the least-melting regions (Fig. 7), peak temperatures remained below the melting point of aluminum, explaining the IMC-free or very thin IMC regions (<3 μm) seen in Fig. 5(a-b). Despite higher thermal conductivity of aluminum, the steel side exhibited steeper gradients due to its higher thermal mass and heat sink behavior. The aluminum flyer, acting as the kinetic energy carrier, experienced rapid adiabatic heating, generating localized temperature spikes even in minimally affected areas [11], [43].


(a) Flyer thickness: 6.35 mm
Al
Steel
1 cm
Al side
Steel side
Al melting point
Steel melting point
Distance (μm)
Temperature (K)
(b) Flyer thickness: 9.53 mm
Al
Steel
1 cm
Al side
Steel side
Al melting point
Steel melting point
Distance (μm)
Temperature (K)
(c) Flyer thickness: 22.86 mm
Al
Al side
Al melting point

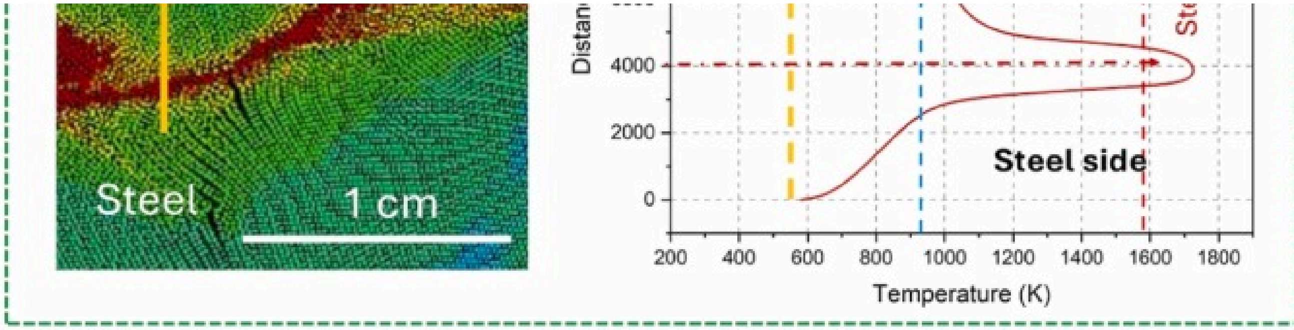


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Fig. 7. EXW AA1100–1018 steel temperature distribution in least-melting region for (a) 6.35 mm flyer at 750 m/s, (b) 9.53 mm at 670 m/s, (c) 22.86 mm at 400 m/s.

Integrating thermal and morphological results establishes a direct process-structure linkage. Melting regions (Fig. 6) align with continuous IMCs at crests, while least-melting regions (Fig. 7) correspond to solid-state bonding at troughs. Increasing flyer thickness led to broader melting layers (1300–3100 µm), which paralleled the thickening of IMCs (3–30 µm). This confirms that flyer thickness governs IMC formation primarily through localized thermal heterogeneity, not wave amplitude alone. Process optimization should therefore focus on controlling both flyer mass and velocity to balance wave morphology with minimal IMC growth.

4.4. Pressure wave propagation and stress wave reflection in EXW

As EXW uses a layer of explosives, which consistently moves the point of detonation, the process provides lasting pressure until the process ends, as shown in Fig. 8(e-g). Hence, the amount of metal forming associated with the flyer movement is relatively consistent, which helps maintain the collision angle, as shown in Fig. 9(a) [11], [41], [43]. This results in the weld having a consistent interface structure over the entire length (Fig. 5)) [13]. The collision angle is significantly higher at the initial and final stage of the EXW process close to the free ends of the workpiece, and their corresponding collision pressure is significantly lower, but after that a consistency is maintained, as also evident through the uniform size of the stress wave extending until the very end of the workpiece length. These observed pressure evolution patterns match prior detonation-driven collision pressure analyses [22], [36], where plasma pressure models predict similar localized peaks and rapid spatial decay at the collision point. Additionally, the initial contact point between the workpiece is not adjacent to the detonation point at a certain distance, causing the end effect during the EXW process, as also shown by Sun et al. [13]. Furthermore, Fig. 10 shows the interfacial temperature and jetting velocity.

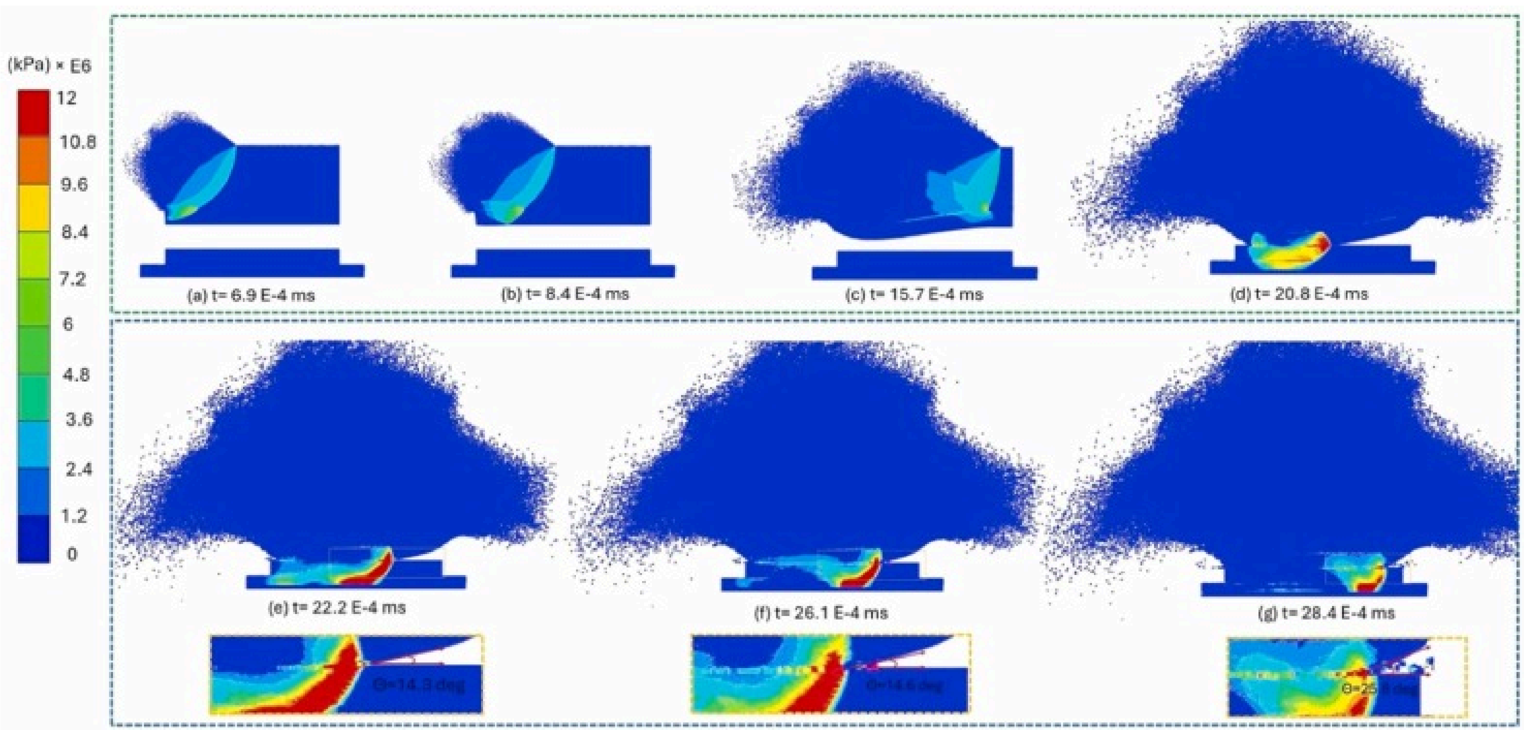

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Fig. 8. Collision pressure distribution during EXW welding of 6.35 mm flyer plate at various time frames (a-g).

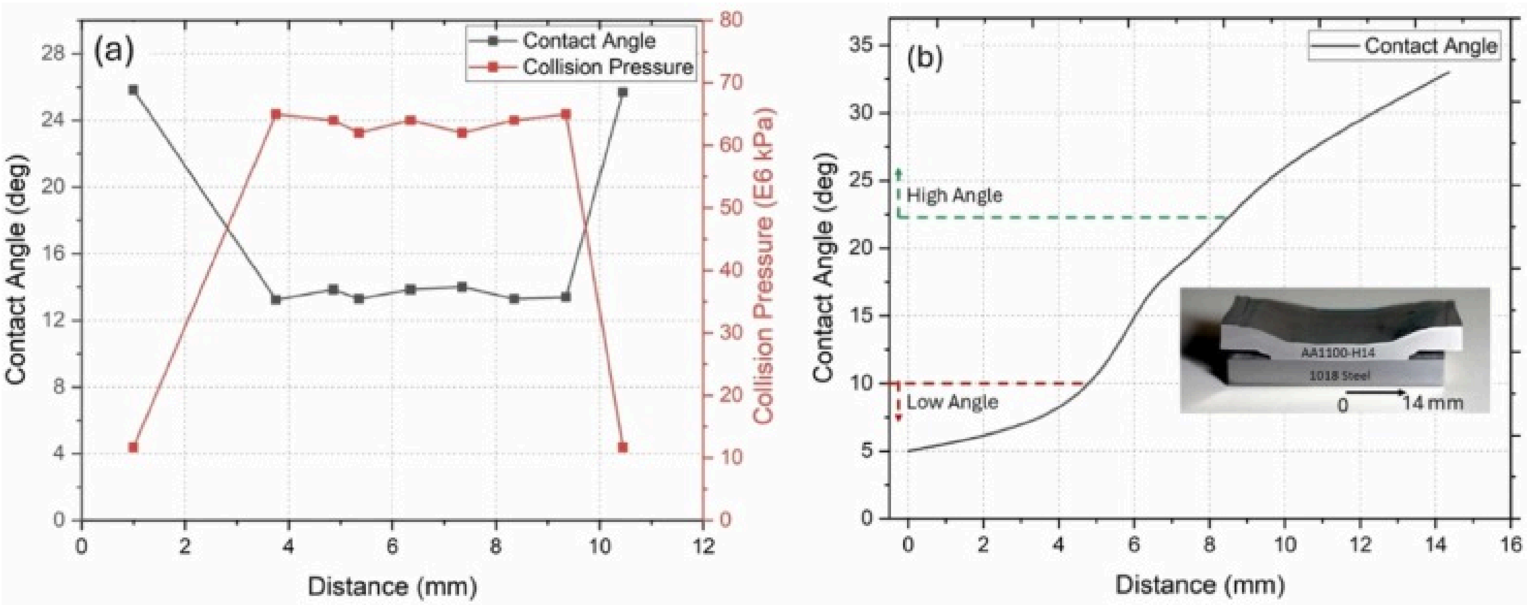

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Fig. 9. Variation of contact angle and collision pressure as obtained from SPH model for 6.35 mm flyer plate for (a) EXW and (b) VFAW.

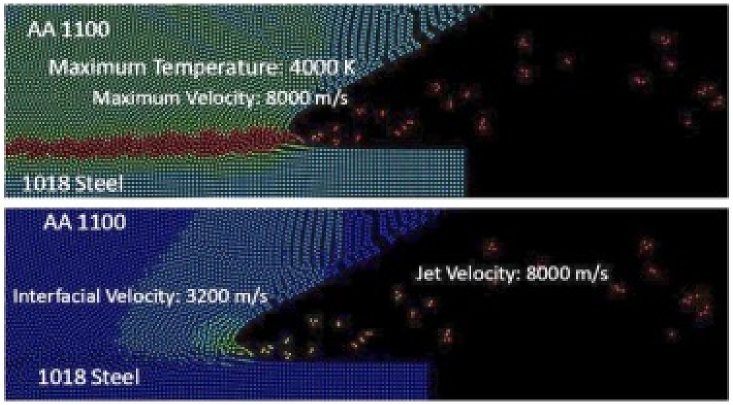
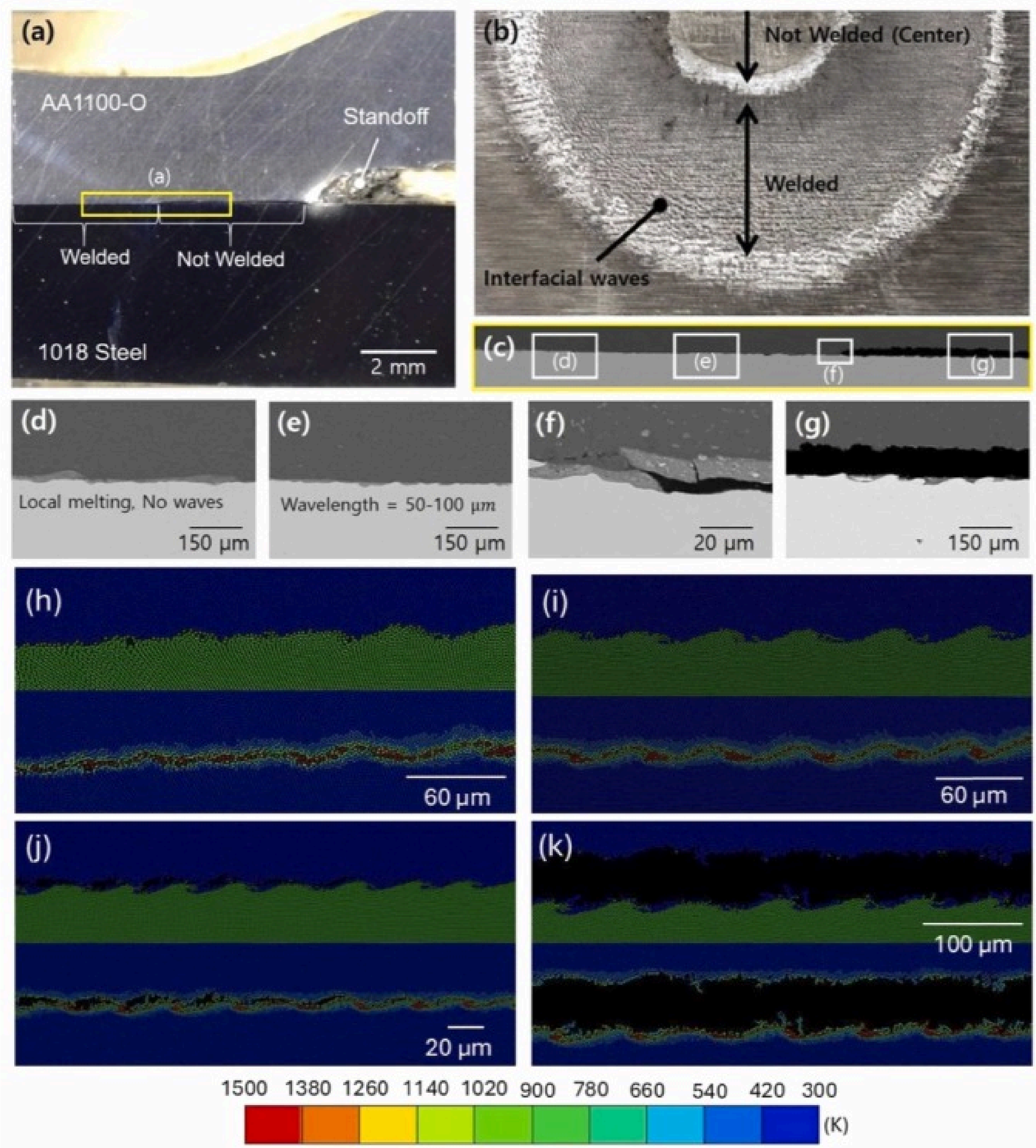


Fig. 10. Temperature and velocity distribution during jetting phenomenon for an EXW welding of a 6.35 mm flyer plate.

4.5. Interface morphology and jetting behavior in VFAW

Unlike EXW, which requires a secure environment and is challenging to implement in small manufacturing settings, VFAW is an agile technology that can be performed in a lab environment [18]. Recent advancements have expanded its capabilities, such as the development of standoff-free VFAW, which simplifies setup and improves process versatility, as demonstrated by Gong et al. [44]. Furthermore, Barnett et al. [45] studied shock-induced effects in VFAW and refined process limits, offering improved predictive models for weld quality. Additional research by Cai et al. [46] highlights its adaptability and potential for broader industrial applications.

Lee et al. [18] reported that aluminum plates up to 8 mm thickness can be successfully joined to steel using VFAW. Based on this, the same 6.35 mm thick flyer used in EXW trials was selected for direct comparison. The resulting interface structures are shown in Fig. 11. Unlike the consistent wavy morphology of EXW interface structures (Fig. 5), VFAW interface structures are non-consistent (heterogeneous) across the welding length. Contrary to the consistent pressure generation in EXW, VFAW utilizes a point-source nature of pressure generation. The flyer velocity and collision angle vary as a function of radial distance from the center, resulting in a continuously evolving interaction geometry, as shown in Fig. 9(b) and Fig. 11.

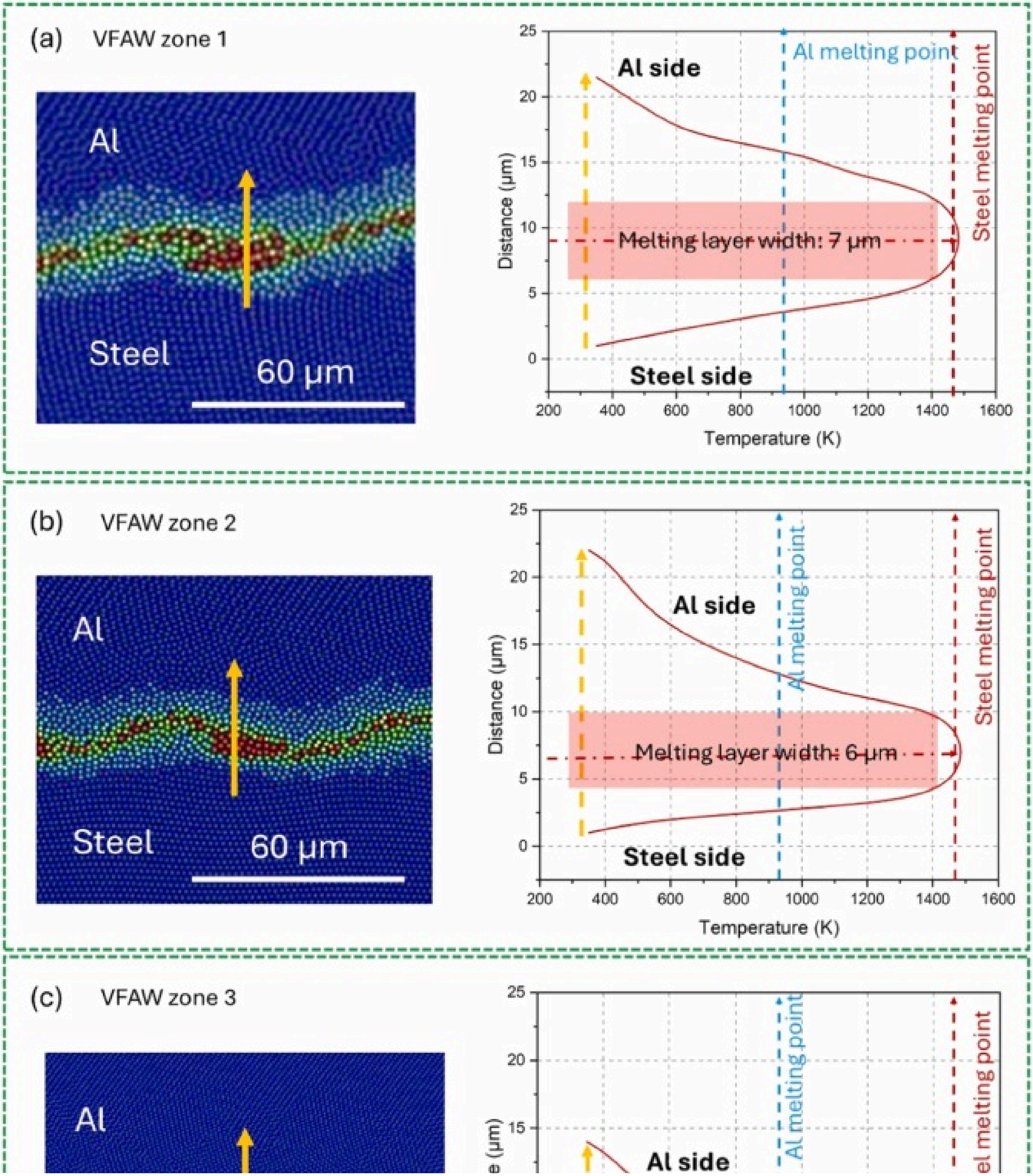


(a)
AA1100-O
Standoff
(a)
Welded
Not Welded
1018 Steel
2 mm
(b)
Not Welded (Center)
Welded
Interfacial waves
(c)
(d)
(e)
(f)
(g)
(d)
Local melting, No waves
150 μm
(e)
Wavelength = 50-100 μm
150 μm
(f)
20 μm
(g)
150 μm
(h)
60 μm
(i)
60 μm
(j)
20 μm
(k)
100 μm
1500 1380 1260 1140 1020 900 780 660 540 420 300
(K)

Fig. 11. VFAW AA1100–1018 steel with 6.35 mm flyer at 660 m/s showing (a) a cross-sectional interface, (b) a fractured surface of the flyer showing the interfacial waves, (c-e) SEM images of the weld interface at various regions and their corresponding SPH model validation interfaces and their respective temperature distribution (h-k).

The interface reveals a spatial progression of bonding regimes from the flyer center outward (Fig. 11 (a-g)), while the overall mechanical properties of the joint are reliable [18]. The interface consists of an unwelded zone due to an insufficient collision angle close to the center (Fig. 11 (b)). Moving radially outward, a melting-dominated region appears, marked by thin (~3–5 µm) intermetallics at crest-like features (Fig. 11 (d)). This transitions into a localized wavy interface where moderate jetting and oblique impact promote solid-state bonding (Fig. 11 (e)), followed by unwelded zone again due to low impact velocity and excessively shallow collision angles at the outer edge (Fig. 11 (g)) [47].

To clarify the spatial heterogeneity observed in the VFAW weld interface (Fig. 11), SPH-predicted temperature distributions were extracted from three distinct zones along the radial direction, corresponding to Fig. 12 (crest regions) and Fig. 13 (troughs). All data represent the same joint fabricated with a 6.35 mm AA1100 flyer at 660 m/s impact velocity. In the melting-dominated zone (zone 1, Fig. 12 (a)), peak temperatures exceeded 750 °C, and a melting layer width of ~7 µm was observed. As one moves outward to zone 2 and zone 3 (Fig. 12 (b-c)), melting layer widths decreased progressively to ~6 µm and ~4 µm, respectively. These trends reflect a radial decay in kinetic energy delivery and collision angle, as captured in Fig. 14 and consistent with the bonding regime progression in Fig. 11.


(a) VFAW zone 1
Al
Steel
60 μm
Al side
Al melting point
Steel melting point
Melting layer width: 7 μm
Steel side
Distance (μm)
Temperature (K)
(b) VFAW zone 2
Al
Steel
60 μm
Al side
Al melting point
Steel melting point
Melting layer width: 6 μm
Steel side
Distance (μm)
Temperature (K)
(c) VFAW zone 3
Al
Al side
Al melting point

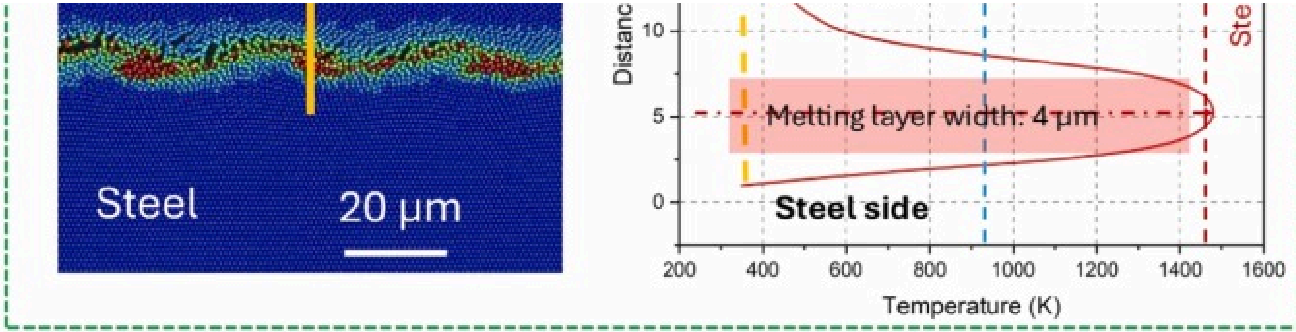


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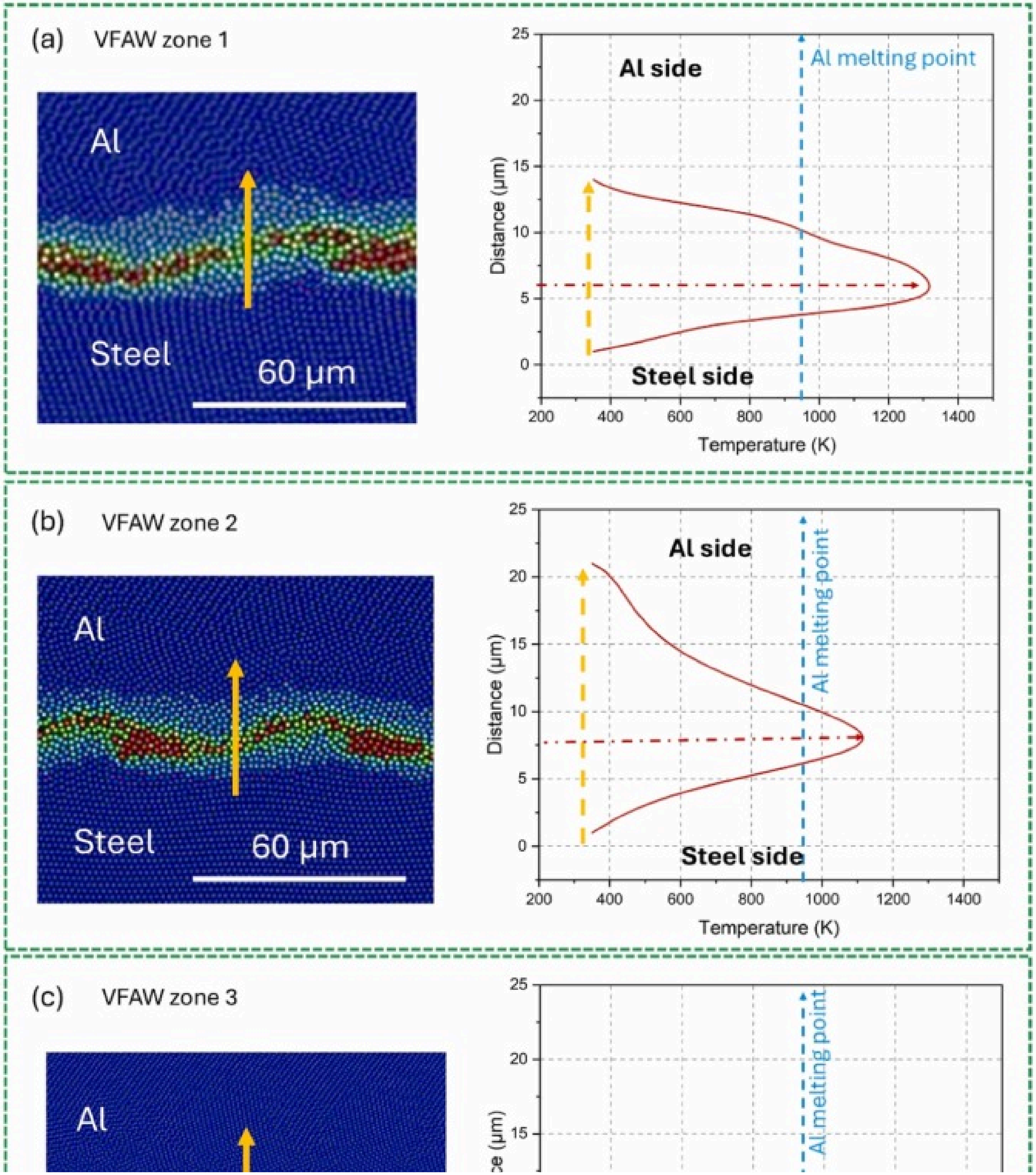
Fig. 12. VFAW AA1100–1018 steel temperature distribution in the melting-dominated region for 6.35 mm flyer at 660 m/s showing (a) zone 1, (b) zone 2, (c) zone 3.


(a) VFAW zone 1
Al
Steel
60 μm
Al side
Al melting point
Steel side
Distance (μm)
Temperature (K)
(b) VFAW zone 2
Al
Steel
60 μm
Al side
Al melting point
Steel side
Distance (μm)
Temperature (K)
(c) VFAW zone 3
Al
Al melting point

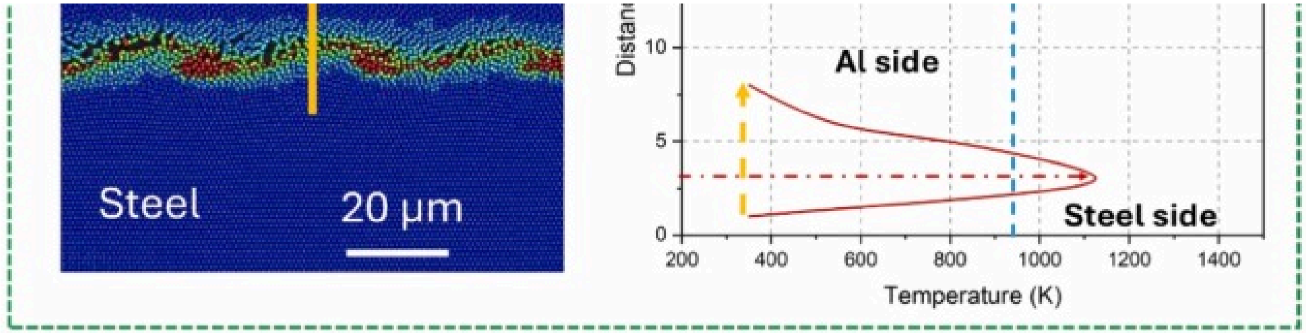


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Fig. 13. VFAW AA1100–1018 steel temperature distribution in the least melting region for 6.35 mm flyer at 660 m/s showing (a) zone 1, (b) zone 2, (c) zone 3.

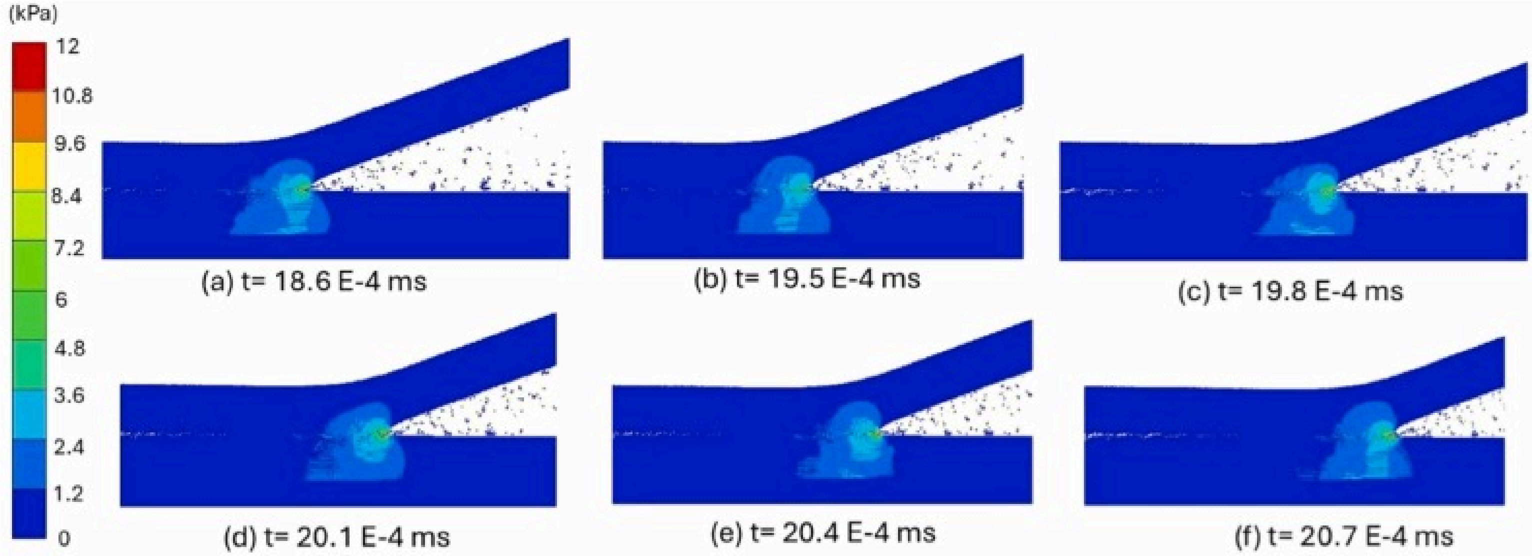

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Fig. 14. Collision pressure distribution during VFAW welding of 6.35 mm flyer plate at various time frames.

In contrast, the trough regions analyzed in Fig. 13 show significantly lower peak temperatures, all remaining below the melting point of aluminum. This supports the observed transition from IMC-rich bonding near crests (Fig. 11 (d)) to solid-state or unbonded regions at troughs (Fig. 11 (c) and Fig. 11 (f)). While EXW exhibits spatial heterogeneity due to detonation timing and wave interactions, heterogeneity in VFAW arises from its point-source pressure application and resulting radial gradient in flyer dynamics [11], [47].

Overall, this dual-region thermal analysis confirms that localized temperature variations, not global impact conditions, govern IMC morphology in VFAW. As in EXW, the aluminum side experiences localized adiabatic heating due to kinetic energy conversion, while the steel acts as a thermal sink, resulting in sharper gradients [43]. The radial pattern in melting behavior and interfacial temperature thus validates the spatially evolving bonding regimes observed experimentally.

Furthermore, the SPH model also showed the reflection of stress wave from the surface of the target and reconnecting into the interface Fig. 14 (a-f), and the phenomenon is observed to be repeated continuously along with the waviness formation at the interface, which aligns with the findings of Godunov et al. [39]. Despite its higher peak pressure, EXW produces longer interfacial wavelengths than VFAW. This counterintuitive trend can be attributed to differences in pressure application characteristics. In EXW, the pressure pulse is sustained over a longer duration and applied across a broader area due to continuous detonation (Fig. 8), allowing more time for stress waves to interact and form extended wave morphologies. In contrast, VFAW delivers a highly localized and short-duration pressure pulse (Fig. 14), leading to more frequent stress wave reflections and shorter interfacial wavelengths. This behavior is consistent with prior studies showing that higher-frequency wave interactions, driven by intense but short-lived pressure, yield smaller wavelength features [25], [26]. Such inverse scaling between peak pressure and wavelength has been documented in prior high-velocity impact welding studies [13], [28], [36], where waveform width and decay rate, in addition to peak magnitude, were shown to influence morphology development.

4.6. Velocity evolution and process parameter effects in VFAW

Fig. 15 presents the velocity profiles of AA1100 flyers recorded using PDV during VFAW trials. In Fig. 15 (a), flyers of 0.5 mm and 1.0 mm thicknesses were launched using identical input energy (5 kJ) to assess how mass influences velocity evolution. The thinner flyer reached a higher final velocity (~650 m/s vs. ~405 m/s) due to its lower inertia [25]. This highlights that even under identical energy conditions, flyer displacement is governed by coupled mass-force dynamics, including losses from plastic deformation and joule heating.

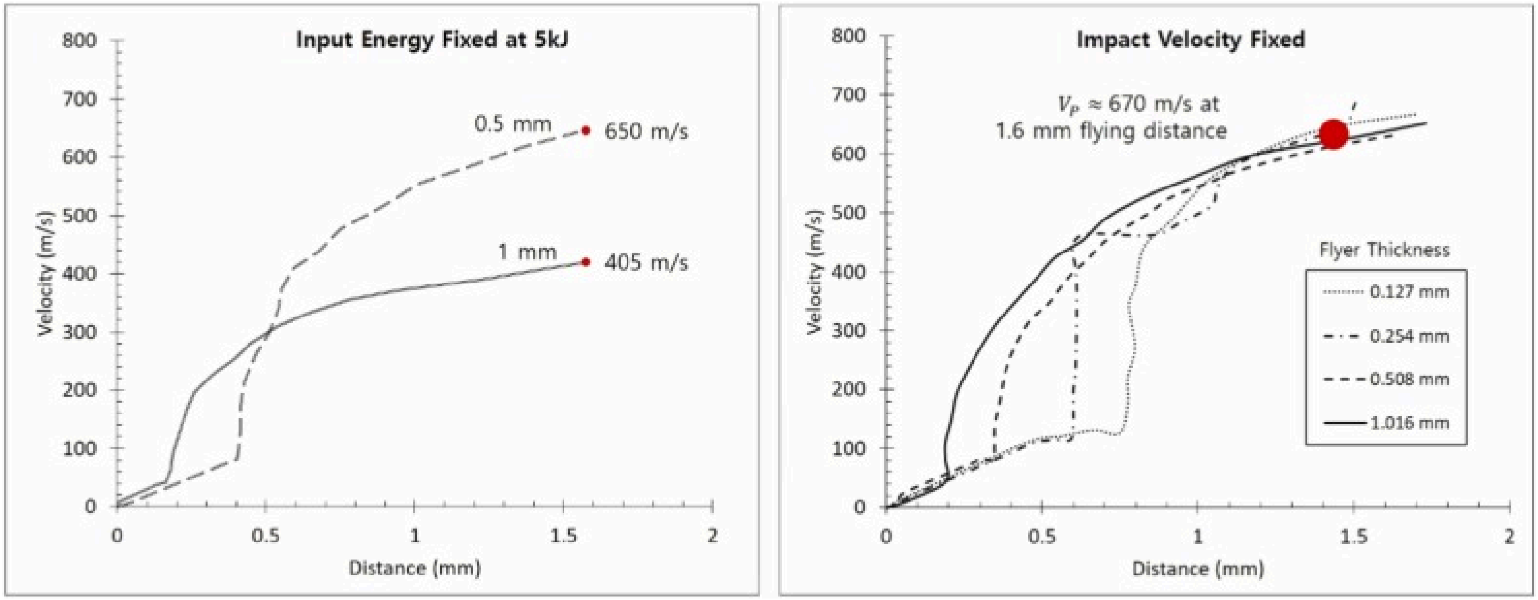

Fig. 15. VFAW AA1100 flyer velocity profiles with distance in the direction of during free flight without target for (a) two flyer thicknesses at 5 kJ input energy, (b) varied thicknesses at constant 670 m/s impact velocity.

Furthermore, to decouple velocity effects from flyer thickness, additional tests were conducted where the input energy was varied to ensure a constant impact velocity (~670 m/s) for four flyer thicknesses: 0.127, 0.254, 0.508, and 1.016 mm (Fig. 15 (b)). All velocity profiles showed a steep acceleration near the initial stage of motion, followed by asymptotic saturation, with thinner flyers accelerating faster and reaching terminal velocity sooner. These experimental insights confirm that flyer mass not only alters peak velocity under constant energy but also modifies the pressure pulse duration and rise time, which are critical for jetting and interfacial heating. While in earlier studies, Lee et al. [25] correlated thicker flyers with larger interfacial waves at fixed velocity, however, the authors did not examine how differences in pressure evolution or energy dissipation affect jetting dynamics.

4.7. Flyer velocity and interface formation in LIW

For the LIW experiment, a varying thickness of AA1100 foil flyers was launched by a laser pulse, and the velocity of each flyer was measured by PDV (Fig. 16 (a)). Considering the standoff distance during LIW equal to flyer thickness, the impact velocity of 0.07 mm, 0.10 mm, 0.13 mm, and 0.15 mm thick flyers were 750 m/s, 660 m/s, 520 m/s and 510 m/s, respectively. In this scenario, the input energy was kept constant to be 3 J for all thicknesses. The velocity profiles show that the flyers accelerate rapidly during the initial 1.5 microseconds and then decelerate until the flyer reaches the standoff distance.

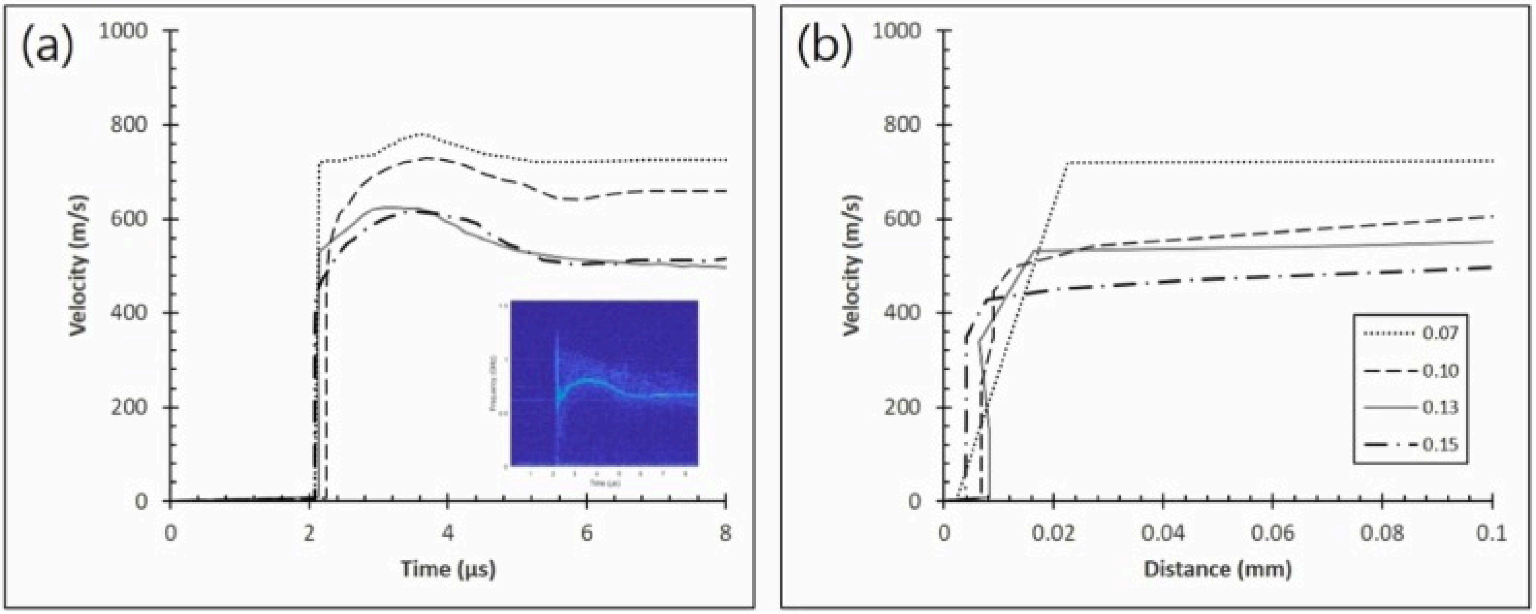

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Fig. 16. LIW AA1100 foil flyer velocity profiles at 3 kJ showing (a) temporal variation with PDV spectrogram, (b) variation with distance (along standoff distance).

4.8. Comparative analysis of interfacial morphology and jetting phenomena

For comparison, a 0.1 mm thick AA1100 foil flyer was welded to 1018 steel by both VFAW and LIW, and the interface structure of the wavy weld interfaces is shown in Fig. 17. While the shape of the wave looks similar, the wavelength of the welds by LIW and VFAW were up to 5 μm and 20 μm and 8 μm and 25 μm during experimental and SPH model observation (Fig. 17), respectively. This is notable because the process seems to affect the size of the interfacial waves even when the collision conditions, such as impact velocity, collision angle, and flyer mass, are kept constant. As LIW was only limited to a 2.5 mm spot size, the standoff distance required (0.1 mm) was much smaller than that of VFAW (1.6 mm).

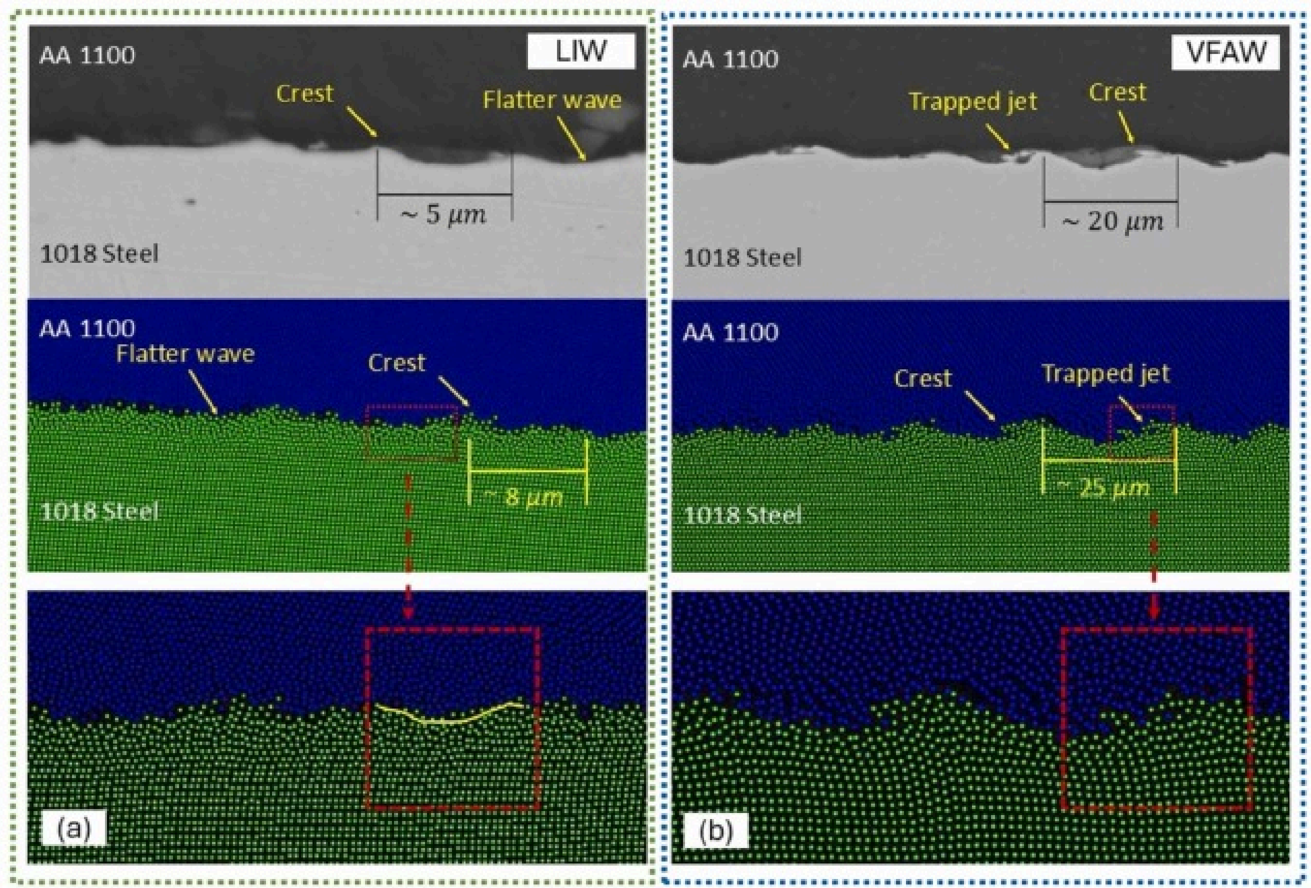

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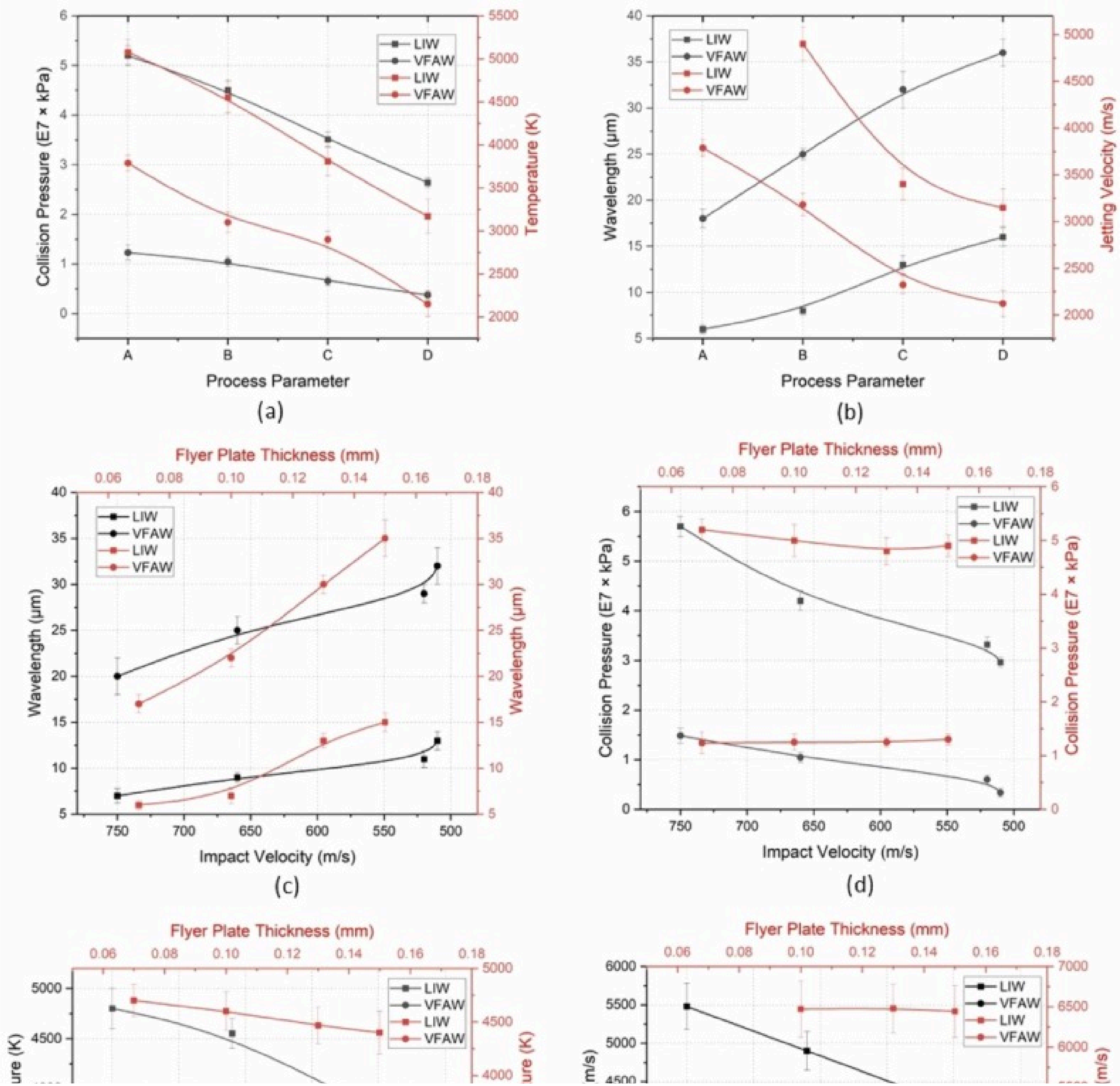
Fig. 17. Comparison of interface structure obtained during experiment and SPH produced by (a) LIW and (b) VFAW for a 0.1 mm thick AA1100 foil flyer launched at 660 m/s to the target 1018 steel.

To further quantify the effect of processes on waviness formation and jetting, the SPH modelling results were elaborated for all the experiments performed for LIW and VFAW at a constant 3 kJ of input energy with the flyer thickness increasing as 0.07 mm, 0.10 mm, 0.13 mm, and 0.15 mm and their corresponding impact velocity decreasing as 750 m/s, 660 m/s, 520 m/s and 510 m/s, respectively represented as process parameter A, B, C and D, respectively. As shown in Fig. 18 (a), both collision pressure and interfacial temperature are observed to decrease progressively from parameter A to D in both LIW and VFAW. LIW consistently exhibits higher peak pressure than VFAW due to its smaller spot size and higher energy concentration. Fig. 18 (b) shows that interfacial wavelength and jetting velocity both increase with flyer thickness (A→D) for LIW and VFAW. In LIW, the wavelength grows from ~6 µm to ~30 µm (a ~400 % increase), while in VFAW it increases from ~18 µm to ~37 µm (~105 %), corresponding to flyer thicknesses from 0.07 mm to 0.15 mm. This trend is nonlinear and governed by two competing factors, (i) thicker flyers carry more mass and momentum, promoting stronger stress wave interactions and deeper interface deformation, and (ii) thinner flyers achieve higher velocities at the same energy due to lower inertia, which increases wave frequency and reduces wavelength [25], [26], [42], [48]. Furthermore, Fig. 18 (c), plotting wavelength versus flyer thickness and impact velocity, confirms that wave growth results from a coupled interplay of flyer momentum and collision conditions. Even though higher velocities are typically associated with finer waves, the dominant factor appears to be flyer mass and resulting pressure pulse duration, especially in VFAW, where broader energy deposition is observed to be enhancing waviness.


Collision Pressure (E7 × kPa)
Temperature (K)
LIW
VFAW
LIW
VFAW
Process Parameter
A
B
C
D
(a)
Wavelength (μm)
Jetting Velocity (m/s)
LIW
VFAW
LIW
VFAW
Process Parameter
(b)
Flyer Plate Thickness (mm)
Wavelength (μm)
Wavelength (μm)
LIW
VFAW
LIW
VFAW
Impact Velocity (m/s)
(c)
Flyer Plate Thickness (mm)
Collision Pressure (E7 × kPa)
Collision Pressure (E7 × kPa)
LIW
VFAW
LIW
VFAW
Impact Velocity (m/s)
(d)
Flyer Plate Thickness (mm)
LIW
VFAW
LIW
VFAW
Flyer Plate Thickness (mm)
LIW
VFAW
LIW
VFAW

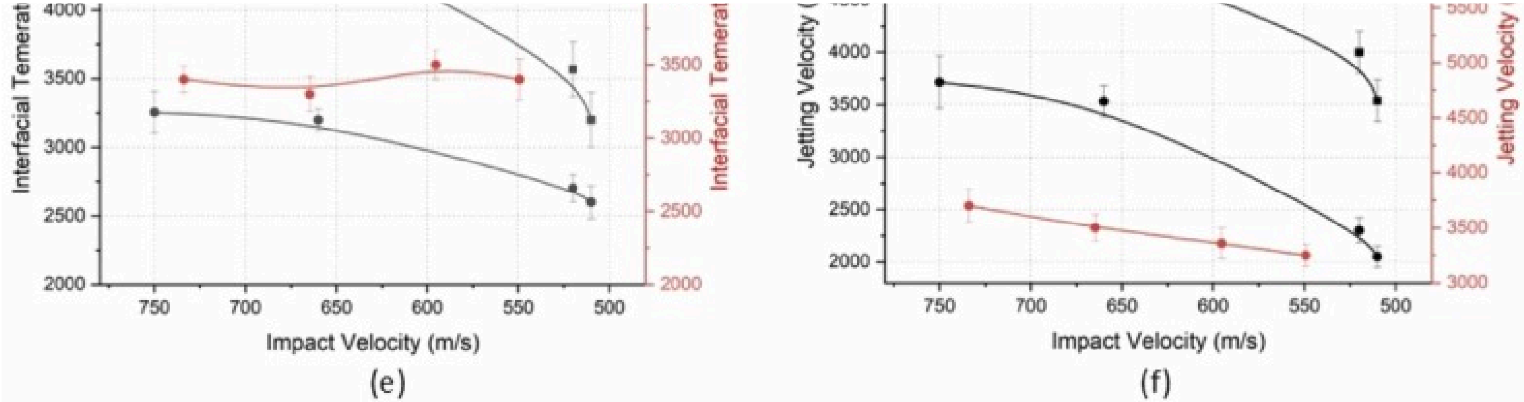


Fig. 18. LIW and VFAW joints (a) collision pressure vs parameters, (b) interfacial temperature vs parameters, (c) wavelength vs impact velocity and flyer thickness, (d) collision pressure vs impact velocity and flyer thickness, (e) interfacial temperature vs impact velocity and flyer thickness, (f) jetting velocity vs impact velocity and flyer thickness.

The fluid behavior of metals during collision is governed not only by peak pressure, but also by the duration and spatial distribution of the pressure pulse [10]. As shown in Fig. 18 (d), as the flyer thickness increased from 0.07 mm to 0.15 mm, and the impact velocity decreased accordingly, the pressure amplitude declined in both processes. In LIW, peak pressure decreased from 5.2×10^7 kPa to 2.64×10^7 kPa, while for VFAW, it decreased from 1.23×10^7 kPa to 3.77×10^6 kPa. This 76 % pressure drop from LIW to VFAW is attributed to an increase in spot size, contributing to lower stress wave intensity and formation of larger wavelengths, also shown in Fig. 19 [36], [37], [38]. This reduction in pressure was also reflected in temperature measurements at the interface, which decreased by 37 % for LIW and by 36 % for VFAW, as shown in Fig. 18 (e) and Fig. 20.

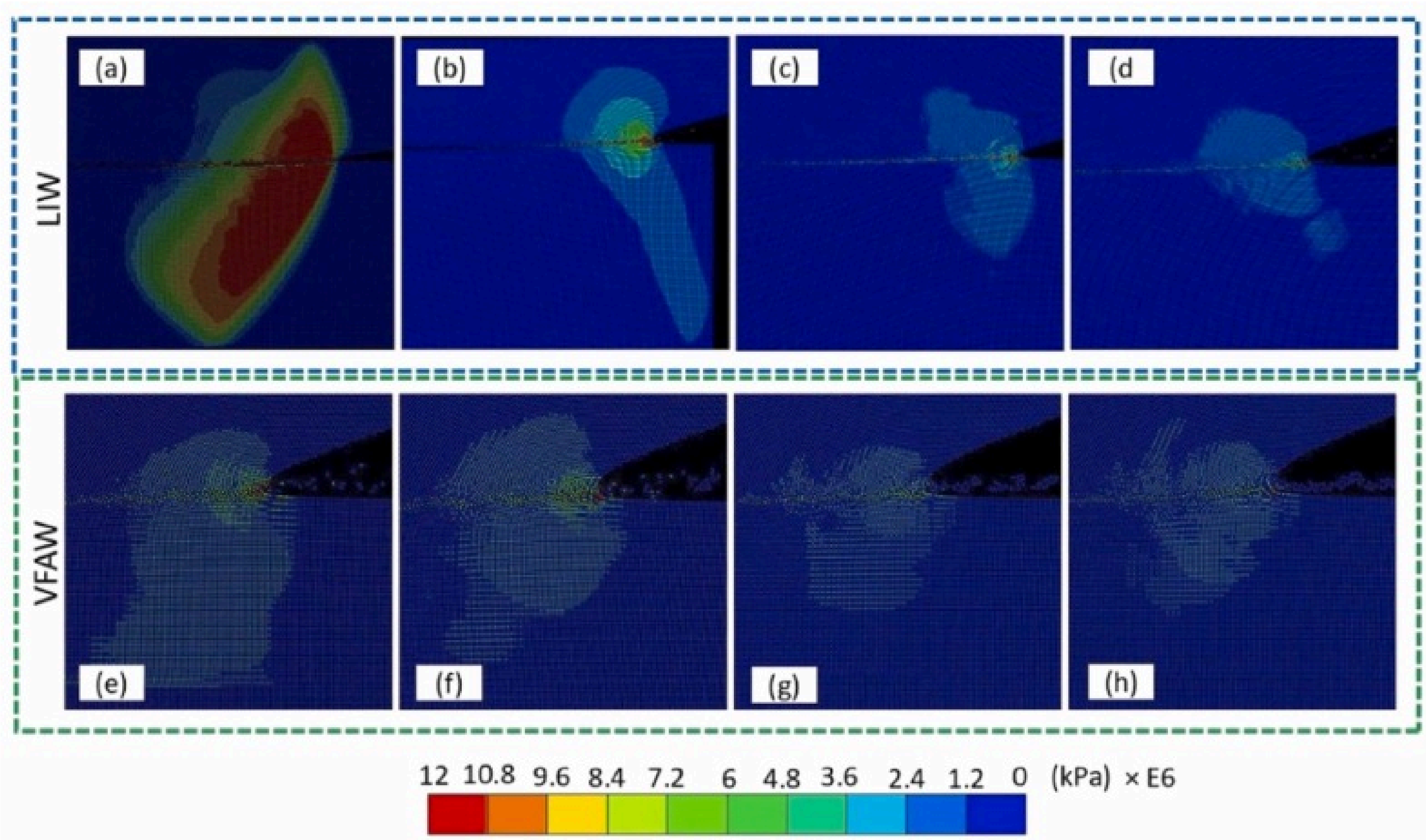


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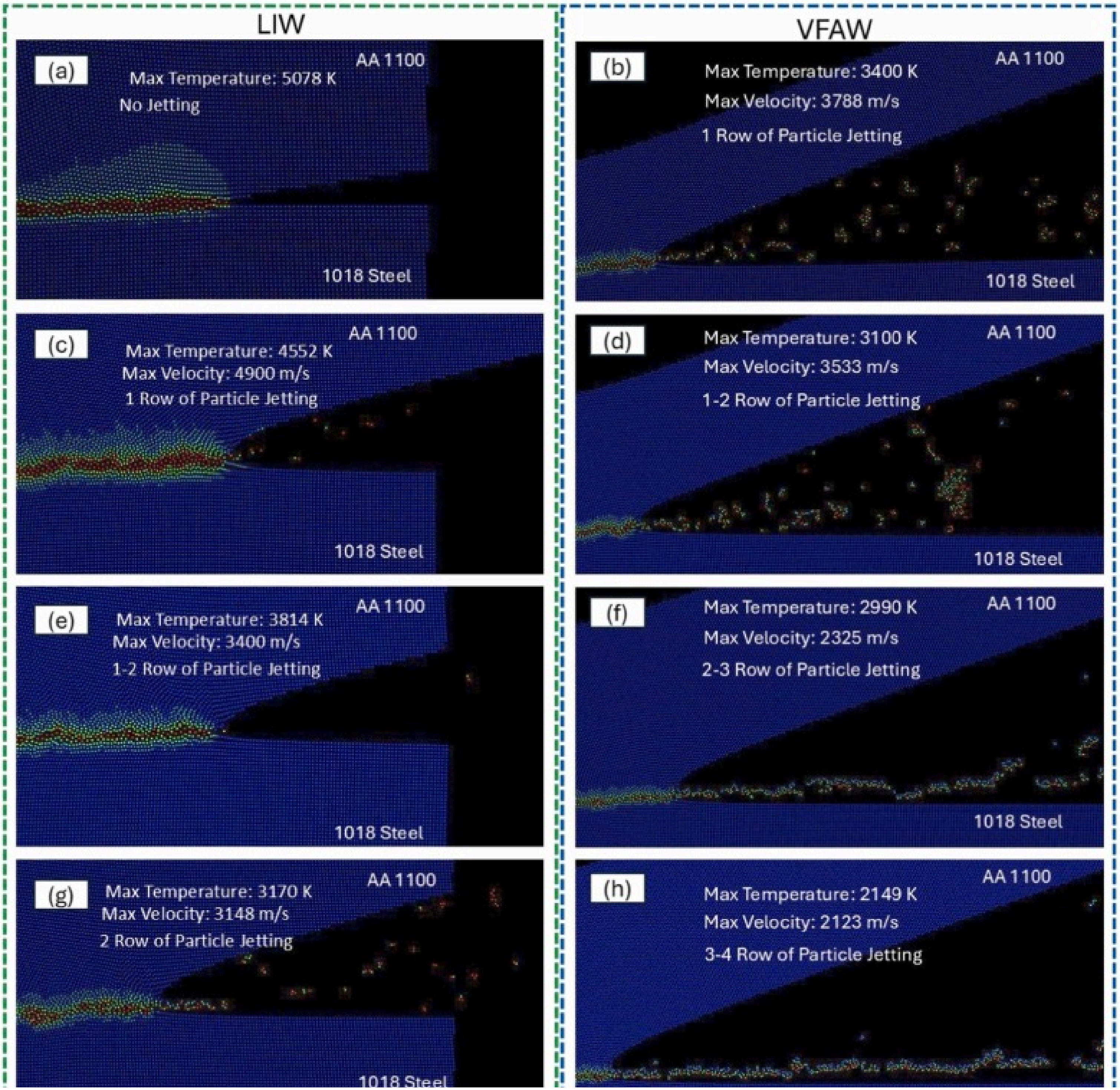
Fig. 19. Collision pressure distribution for LIW and VFAW at parameters A, B, C and D in the panels (a-d) and (e-f), respectively.


LIW
VFAW
(a)
Max Temperature: 5078 K
No Jetting
AA 1100
1018 Steel
(b)
Max Temperature: 3400 K
Max Velocity: 3788 m/s
1 Row of Particle Jetting
AA 1100
1018 Steel
(c)
Max Temperature: 4552 K
Max Velocity: 4900 m/s
1 Row of Particle Jetting
AA 1100
1018 Steel
(d)
Max Temperature: 3100 K
Max Velocity: 3533 m/s
1-2 Row of Particle Jetting
AA 1100
1018 Steel
(e)
Max Temperature: 3814 K
Max Velocity: 3400 m/s
1-2 Row of Particle Jetting
AA 1100
1018 Steel
(f)
Max Temperature: 2990 K
Max Velocity: 2325 m/s
2-3 Row of Particle Jetting
AA 1100
1018 Steel
(g)
Max Temperature: 3170 K
Max Velocity: 3148 m/s
2 Row of Particle Jetting
AA 1100
1018 Steel
(h)
Max Temperature: 2149 K
Max Velocity: 2123 m/s
3-4 Row of Particle Jetting
AA 1100

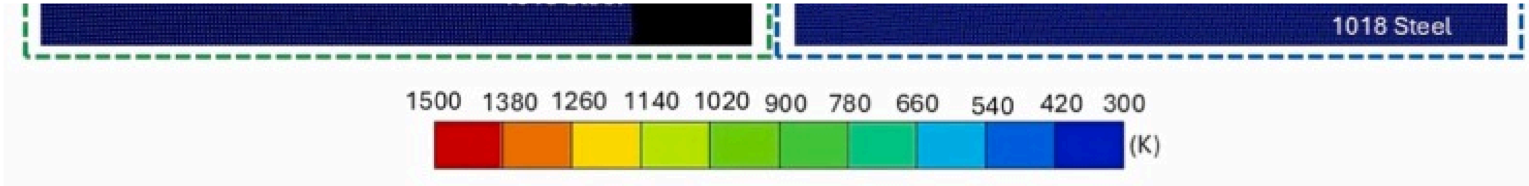


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Fig. 20. Temperature distribution for LIW and VFAW jetting at parameters A, B, C and D in the panels (a), (c), (e), (g) and (b), (d), (f), (h), respectively.

Jetting behavior also exhibited a strong dependence on flyer thickness and impact energy. As shown in Fig. 18 (f) and Fig. 21, jetting particle counts increased with flyer thickness, while jetting velocity decreased due to reduced collision pressure. The acceleration of the jet was driven by high-temperature and high-pressure gases generated during the oblique collision of the flyer plate with the target plate [49]. In LIW, no jetting was observed at the lowest parameter (parameter A) due to intense stress wave interactions and insufficient mass availability. As the thickness increased, the jetting particle number rose from 16 to 89, correlating with greater mass availability. However, jetting velocity decreased from 4900 m/s to 3148 m/s, as shown in Fig. 20, likely due to reduced collision pressure. In contrast, VFAW demonstrated a marked increase in jetting, with particle counts rising from 247 to 633 across parameters, while jetting speeds decreased from 3788 m/s to 2123 m/s, reflecting similar trends in pressure reduction (Fig. 21). The higher jetting activity in VFAW is attributed to its more distributed interaction and broader contact zone, despite its lower peak pressures. Compositional analysis of jetting particles showed that aluminum dominated the jet, with significantly less iron content. This is due to density-based momentum partitioning, the lower-density aluminum is preferentially ejected upon collision, consistent with prior observations [50].

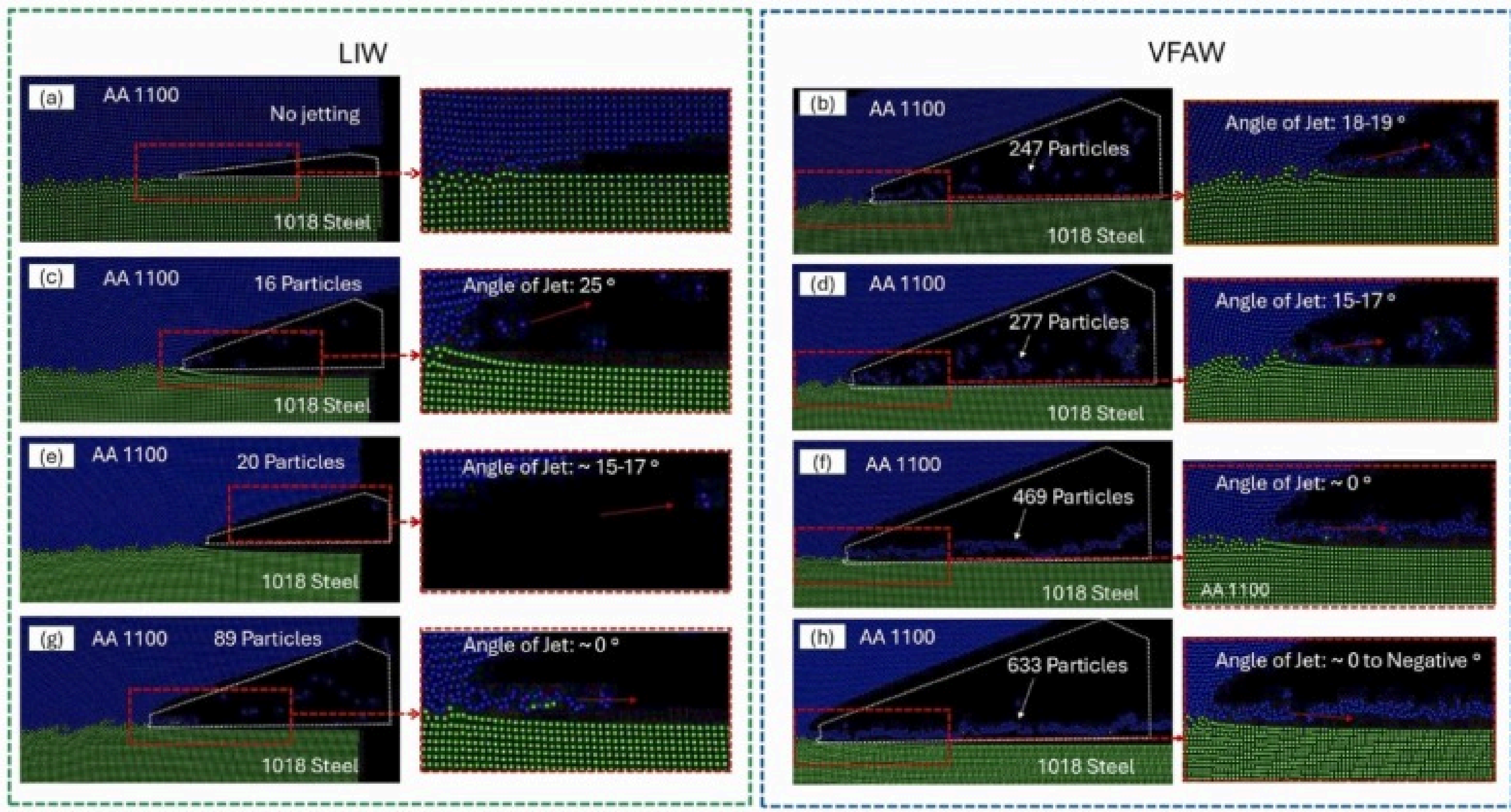

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Fig. 21. Jetting material distribution for LIW and VFAW at parameters A, B, C and D in the panels (a), (c), (e), (f) and (b), (d), (f), (h), respectively.

Fig. 22 provides further insight into the pressure pulse propagation during LIW. SPH simulations of a 0.15 mm flyer showed that the pressure wave reflects from both the flyer and target surfaces and propagates asymmetrically, the bulk of the stress wave penetrates deeper into the denser target, resulting in spring-back [28]. Additionally, the contact angle increases progressively from the weld center to the edge, contributing to heterogeneity in wave formation and jetting distribution.

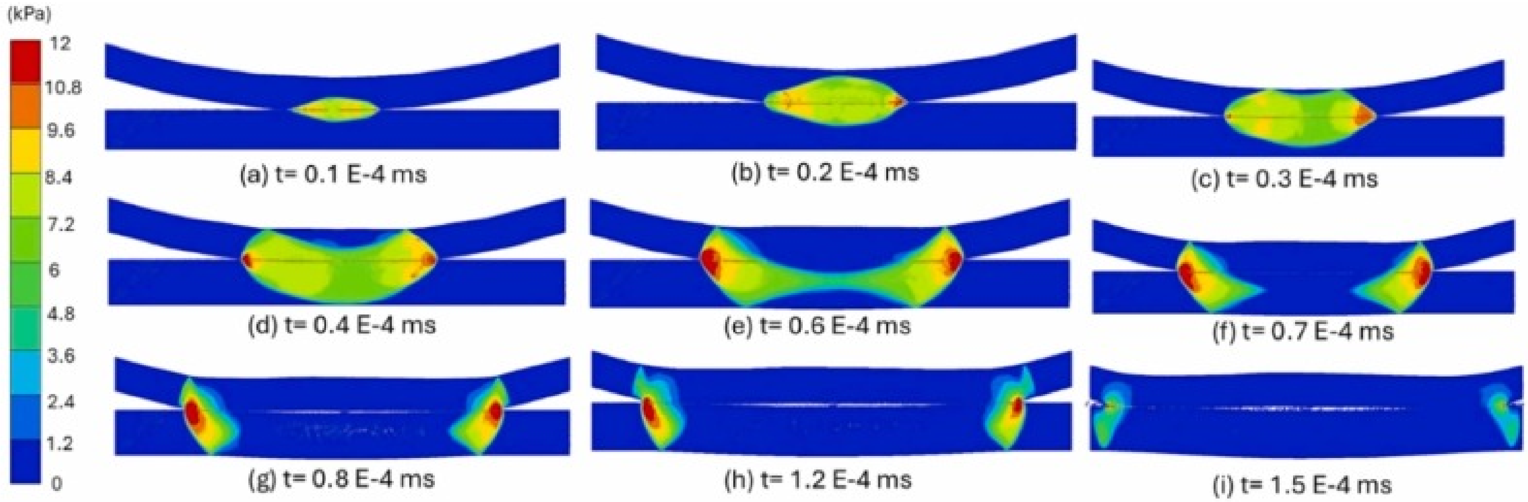

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Fig. 22. Collision pressure distribution during LIW welding of 0.15 mm flyer plate at various time frames.

To simplify the complexities arising from the variation of two parameters (flyer plate thickness and impact velocity) and to better understand their individual effects on interfacial waviness formation and jetting, SPH models were conducted under two distinct scenarios. The first scenario involved varying the impact velocity (750 m/s, 660 m/s, 520 m/s, and 510 m/s) while maintaining a constant flyer plate thickness of 0.1 mm. The second scenario examined the variation of flyer plate thickness (0.07 mm, 0.1 mm, 0.13 mm, and 0.15 mm) at a constant impact velocity of 750 m/s, applicable to both LIW and VFAW.

4.9. Effects of impact velocity variation at constant flyer thickness

In the first scenario, the interfacial wavelength in LIW increased by 100 %, from 7 µm to 14 µm, while VFAW saw a 70 % increase, from 20 µm to 32 µm as shown in Fig. 18 (c). This trend reaffirms that lower impact velocities reduce collision pressure and stress wave intensity, allowing longer wavelength formation [13]. However, even under identical velocity, VFAW consistently exhibited larger wavelengths than LIW, reinforcing the influence of pulse duration and spatial energy distribution. Additionally, in LIW, the pressure at the interface decreased significantly by approximately 48 %, from 5.7×10^7 kPa to 2.962×10^7 kPa. In contrast, VFAW exhibited a 77 % reduction in pressure, from 1.485×10^7 kPa at 750 m/s to 3.37×10^6 kPa at 510 m/s (Fig. 18 (d) and Fig. 23).

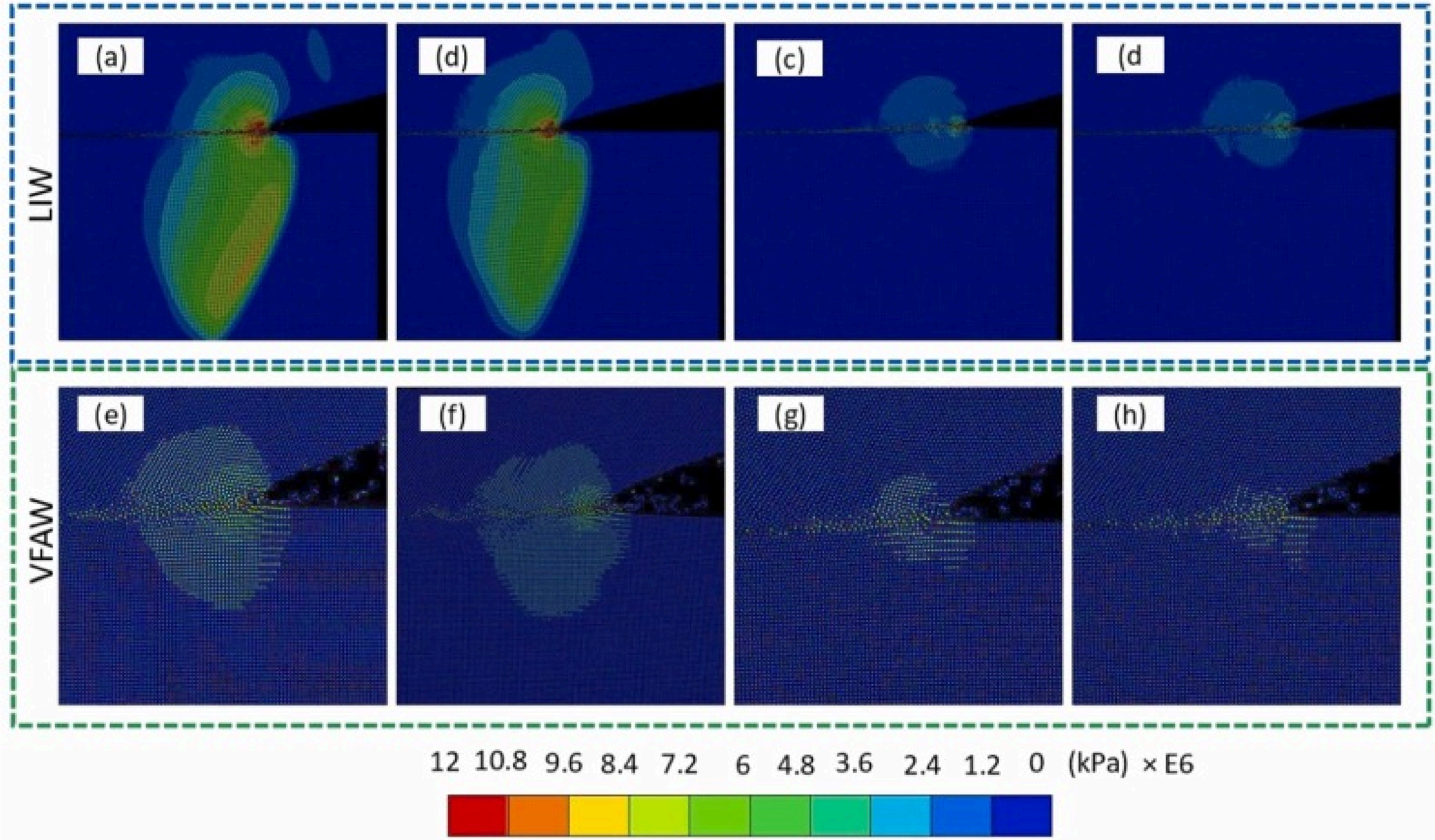

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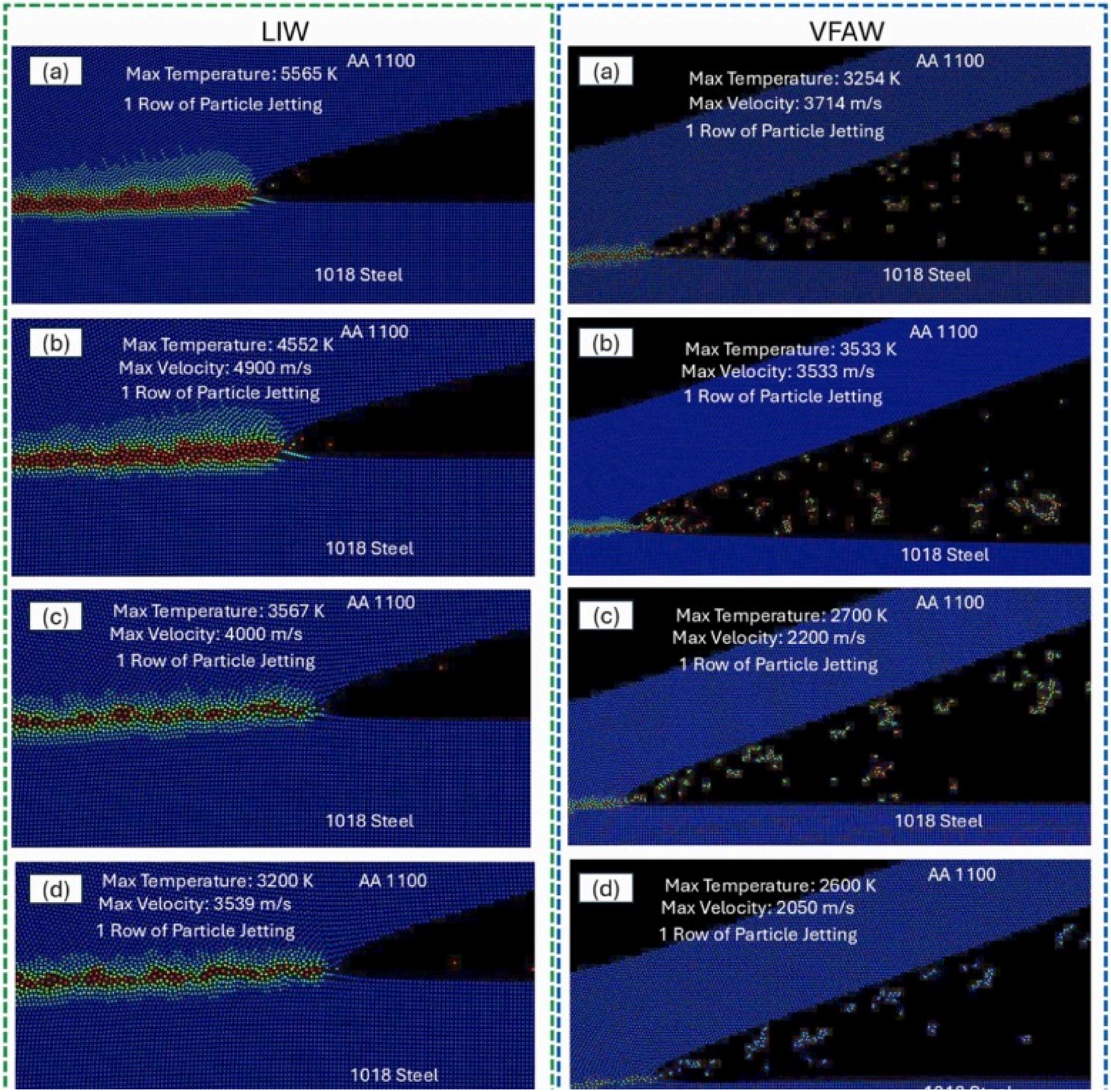
Fig. 23. Collision pressure distribution for LIW and VFAW at 750 m/s, 660 m/s, 520 m/s and 510 m/s in the panels (a-d) and (e-f), respectively.

This indicates that VFAW experiences a more pronounced drop in pressure compared to LIW under similar conditions; however, the stress wave size drops significantly after reflecting from the target plate during LIW (Fig. 19) compared to VFAW as also reflected in higher temperature stability during VFAW (25 % drop, 3254 K to 2600 K) than LIW (42 % drop, 5565 K to 3200 K) (Fig. 24). Additionally, the jetting intensity in LIW decreased by 35 %, from 5479 m/s at 750 m/s impact velocity to 3539 m/s at 510 m/s.

Conversely, VFAW jetting speed dropped by 45 %, from 3714 m/s to 2050 m/s, over the same range of impact velocities (Fig. 25). These differences highlight how the distributed energy input of VFAW supports more sustained wave-jet dynamics, while LIW is more sensitive to changes in velocity due to its highly localized pressure pulse [15], [21].


LIW
(a)
Max Temperature: 5565 K
1 Row of Particle Jetting
AA 1100
1018 Steel
(b)
Max Temperature: 4552 K
Max Velocity: 4900 m/s
1 Row of Particle Jetting
AA 1100
1018 Steel
(c)
Max Temperature: 3567 K
Max Velocity: 4000 m/s
1 Row of Particle Jetting
AA 1100
1018 Steel
(d)
Max Temperature: 3200 K
Max Velocity: 3539 m/s
1 Row of Particle Jetting
AA 1100
1018 Steel
VFAW
(a)
Max Temperature: 3254 K
Max Velocity: 3714 m/s
1 Row of Particle Jetting
AA 1100
1018 Steel
(b)
Max Temperature: 3533 K
Max Velocity: 3533 m/s
1 Row of Particle Jetting
AA 1100
1018 Steel
(c)
Max Temperature: 2700 K
Max Velocity: 2200 m/s
1 Row of Particle Jetting
AA 1100
1018 Steel
(d)
Max Temperature: 2600 K
Max Velocity: 2050 m/s
1 Row of Particle Jetting
AA 1100

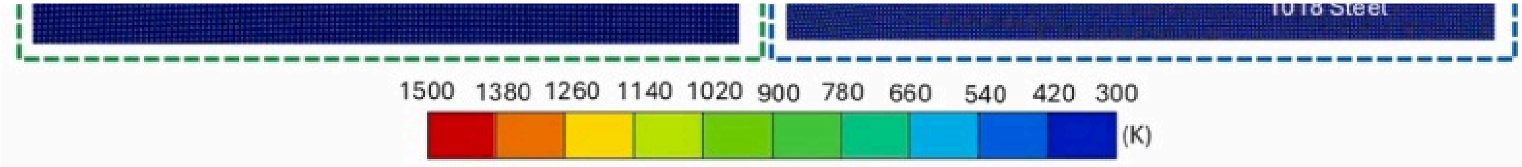


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Fig. 24. Temperature distribution for LIW and VFAW jetting at 750 m/s, 660 m/s, 520 m/s and 510 m/s in the panels (a), (c), (e), (g) and (b), (d), (f), (h), respectively.

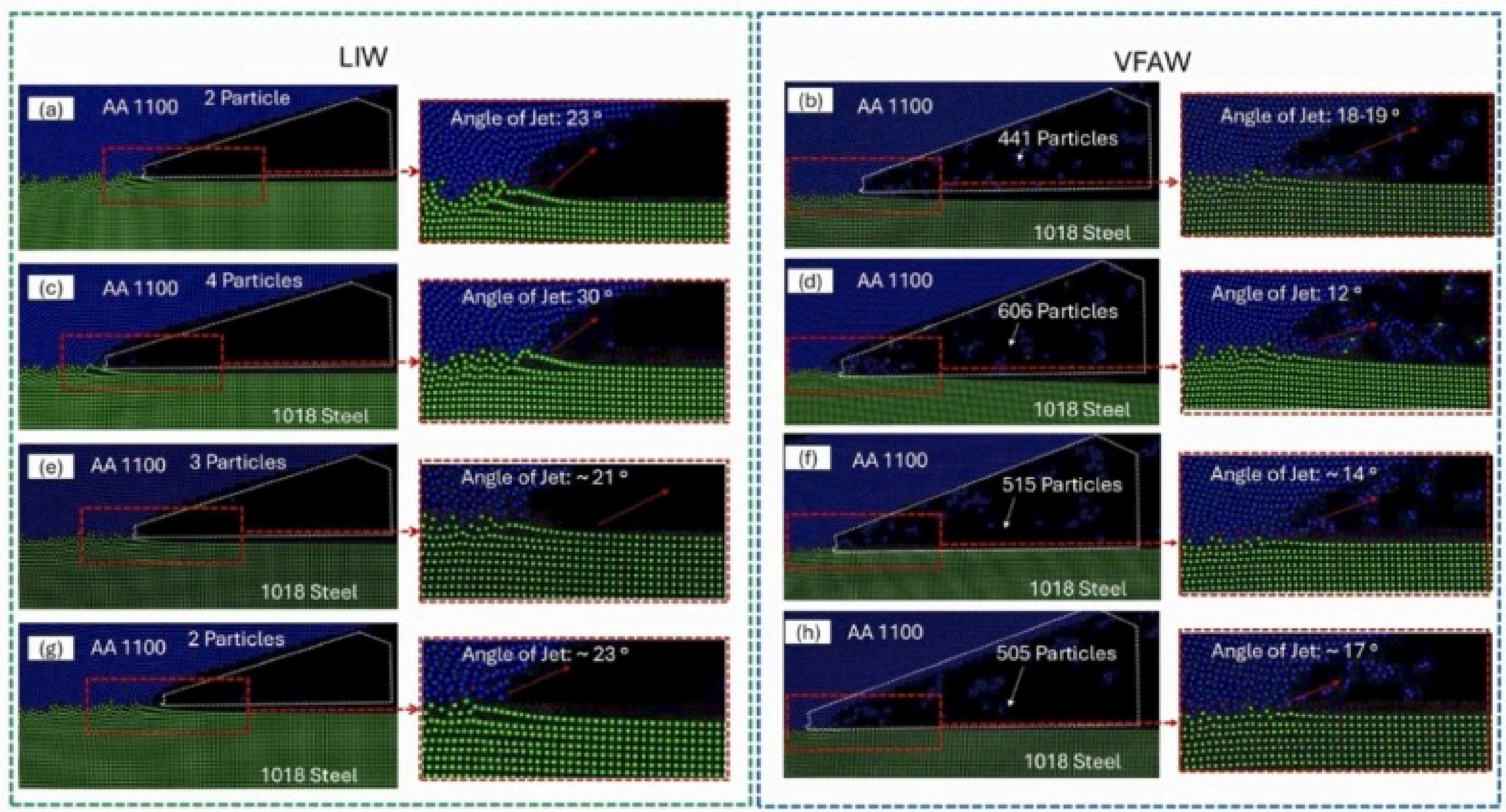

Fig. 25. Jetting material distribution for LIW and VFAW at 750 m/s, 660 m/s, 520 m/s and 510 m/s in the panels (a), (c), (e), (f) and (b), (d), (f), (h), respectively.

4.10. Effects of flyer thickness variation at constant impact velocity

In the second scenario, flyer thickness was varied (0.07 mm to 0.15 mm) while keeping the impact velocity fixed at 750 m/s to isolate the effect of flyer mass on interfacial and jetting behavior in both LIW and VFAW. As shown in Fig. 26, collision pressure remained relatively stable within each process, but the spatial extent of the pressure pulse increased with flyer thickness. This expansion was more pronounced in VFAW due to its larger interaction area and flyer spot size [18]. Temperature profiles followed a similar trend (Fig. 27), while peak interface temperatures remained within a narrow range, the high-temperature region broadened with increasing flyer mass. In LIW, the temperature remained centered but compact, whereas VFAW showed more distributed thermal zones [26], [35]. Despite the near-constant pressure and velocity, interfacial wavelength increased by 105 % from 17 µm to 35 µm as the flyer plate thickness increased by 114 % from 0.07 to 0.15 mm at a constant velocity with the increase in thickness, confirming that flyer thickness alone can promote wave growth via enhanced momentum transfer and prolonged contact. However, it highlighted a decrement of 76 % of collision pressure from 5.2 × 107 kPa to 1.27 × 10^7 kPa, 26 % of interface temperature from 4600 K to 3400 K and 47 % of jetting velocity from 6400 m/s to 3400 m/s between LIW and VFAW, underscoring how pulse duration and localization affect collision dynamics.

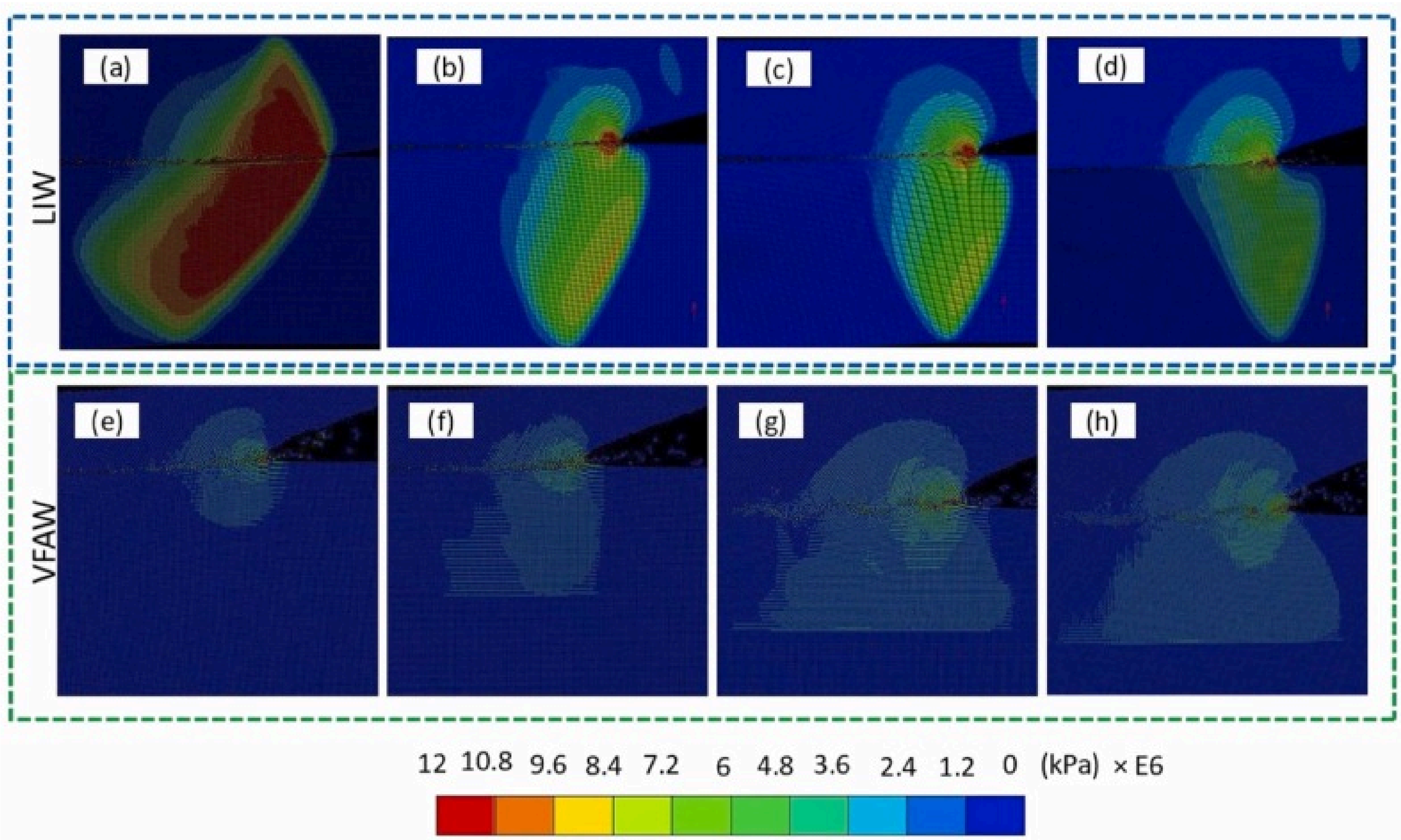

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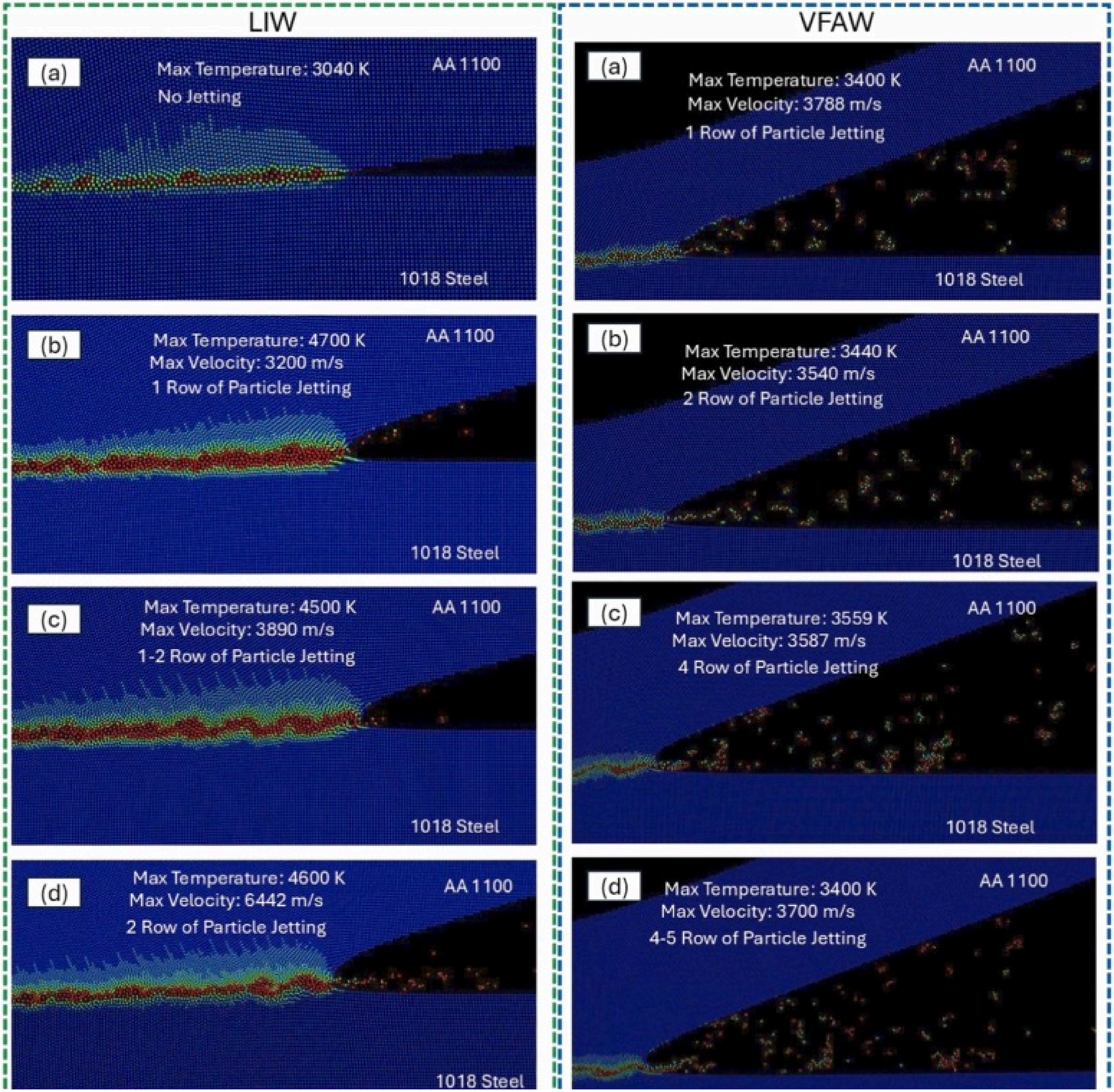
Fig. 26. Collision pressure distribution for LIW and VFAW at 750 m/s for flyer plate thickness of 0.07 mm, 0.1 mm, 0.13 mm and 0.15 mm in the panels (a-d) and (e-f), respectively.


LIW
(a)
Max Temperature: 3040 K
No Jetting
AA 1100
1018 Steel
(b)
Max Temperature: 4700 K
Max Velocity: 3200 m/s
1 Row of Particle Jetting
AA 1100
1018 Steel
(c)
Max Temperature: 4500 K
Max Velocity: 3890 m/s
1-2 Row of Particle Jetting
AA 1100
1018 Steel
(d)
Max Temperature: 4600 K
Max Velocity: 6442 m/s
2 Row of Particle Jetting
AA 1100
1018 Steel
VFAW
(a)
Max Temperature: 3400 K
Max Velocity: 3788 m/s
1 Row of Particle Jetting
AA 1100
1018 Steel
(b)
Max Temperature: 3440 K
Max Velocity: 3540 m/s
2 Row of Particle Jetting
AA 1100
1018 Steel
(c)
Max Temperature: 3559 K
Max Velocity: 3587 m/s
4 Row of Particle Jetting
AA 1100
1018 Steel
(d)
Max Temperature: 3400 K
Max Velocity: 3700 m/s
4-5 Row of Particle Jetting
AA 1100

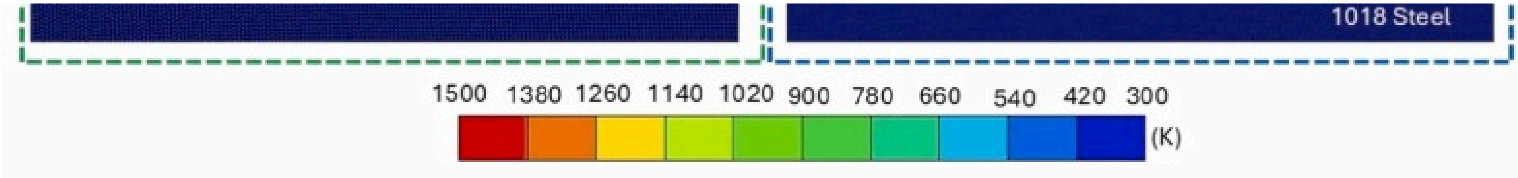


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Fig. 27. Temperature distribution for LIW and VFAW jetting at 750 m/s for flyer plate thickness of 0.07 mm, 0.1 mm, 0.13 mm and 0.15 mm in the panels (a), (c), (e), (g) and (b), (d), (f), (h), respectively.

The jetting quantity, however, has significantly increased as the flyer thickness increases (Fig. 28). For LIW, from no jetting at 0.07 mm of thickness to 10 particles at 0.1 mm, 25 particles at 0.13 mm and 45 particles at 0.15 mm, marking an approximately 400 % increment in jetting since jetting started and by 88 %, from 437 particles to 824 particles for VFAW, reflecting enhanced material ejection with thicker flyers and broader energy coupling.

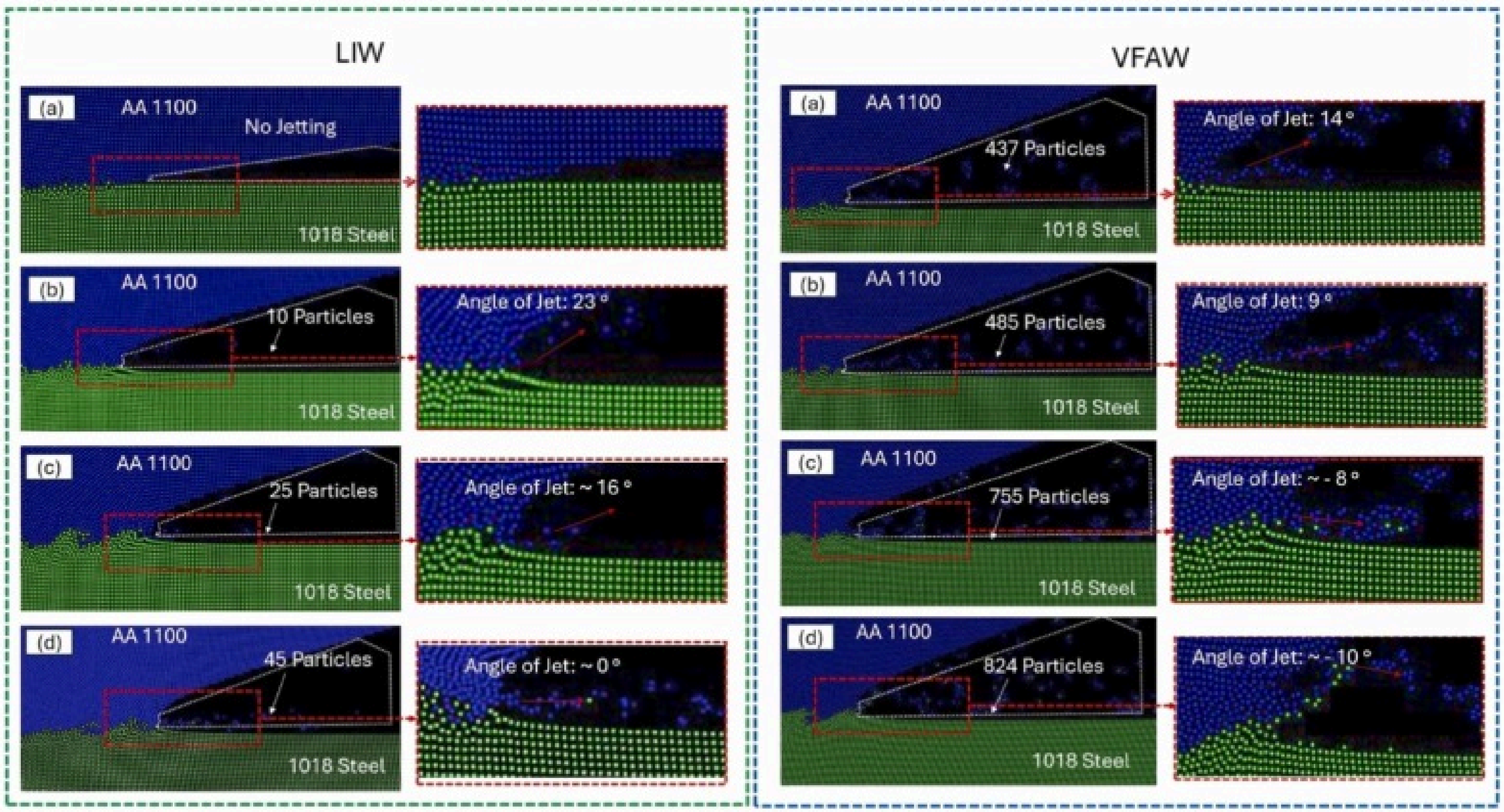

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Fig. 28. Jetting material distribution for LIW and VFAW at 750 m/s for flyer plate thickness of 0.07 mm, 0.1 mm, 0.13 mm and 0.15 mm in the panels (a), (c), (e), (f) and (b), (d), (f), (h), respectively.

These trends reveal a fundamental decoupling between jetting velocity and jetting quantity. Jetting velocity is primarily controlled by peak collision pressure (higher in LIW), while quantity is governed by mass availability and pressure pulse footprint (greater in VFAW and EXW). EXW, with the most prolonged and distributed pulse, produces the most extensive jetting but at lower velocities. VFAW balances between intensity and spatial spread, while LIW yields sharp, high-speed jets but is limited in number. This divergence highlights the importance of distinguishing mass-driven and pressure-driven jetting mechanisms across different high-strain-rate welding methods.

5. Conclusions

This study advances a unified mechanistic framework for predicting interfacial morphology and jetting in high-velocity impact welding. Across EXW, VFAW, and LIW, results show that pressure-pulse geometry, duration and spatial distribution rather than process-specific mechanics alone, governs wave formation, thermal gradients, IMC development, and jetting.

- Wave scaling and morphology envelope: Interfacial wavelengths ranged from ~6–30 µm (LIW) to ~18–37 µm (VFAW) and up to ~2.9 mm (EXW). In EXW, increasing flyer thickness from 6.35 mm to 22.86 mm (velocity 750 → 400 m/s) increased wave amplitude and wavelength, indicating flyer mass/total kinetic energy can outweigh velocity in controlling wave growth.

- Thermal-microstructural linkage: Melting zone width scaled with flyer mass and pulse duration: EXW broadened from ~1.3 mm to ~3.1 mm with IMCs thickening from ~3–5 µm to ~20–30 µm at crests; VFAW produced ~0.4–1.0 mm melting zones with thin, discontinuous IMCs; LIW yielded ~50–200 µm zones with negligible IMCs. Crest–trough asymmetry confirms that thermal heterogeneity, not wave geometry alone, controls IMC continuity.

- Pressure-pulse control: In LIW, peak interface pressure decreased from ~5.2×10^7 to ~2.64×10^7 kPa with flyer thickness; in VFAW, from ~1.23×10^7 to ~3.77×10^6 kPa. Wavelengths lengthened as pressure decreased, with LIW retaining higher pressures due to tighter energy localization.

- Jetting decoupling: Jet quantity scaled with flyer thickness and interaction footprint (EXW > VFAW > LIW), while jet velocity followed peak pressure (LIW > VFAW > EXW), showing mass-dominated quantity versus pressure-dominated velocity.

- Mechanistic unification and design guidance: Longer, broader pulses favor large, continuous waves with greater IMC formation (EXW), short, localized pulses yield finer features with minimal IMCs (LIW), VFAW occupies an intermediate regime. Optimizing flyer mass, standoff/spot size, and pulse duration enables application-specific control, from large-area cladding (EXW) to mid-gauge dissimilar joints (VFAW) to precision, low-IMC bonds (LIW).

CRediT authorship contribution statement

Glenn Daehn: Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Curtis Prothe:** Writing – review & editing, Validation, Resources, Funding acquisition, Conceptualization. **Anupam Vivek:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Data curation. **Deepak Kumar:** Writing – review & editing, Visualization, Investigation, Formal analysis. **Taeseon Lee:** Writing – original draft, Investigation, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Taeseon Lee reports financial support was provided by National Science Foundation. Taeseon Lee reports financial support was provided by Incheon National University. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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